

Master's Thesis

**Last in Translation? Evaluating Language-Invariant
Reason Ranking in Numerical Fact-Checking**

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I hereby declare that I have written this thesis independently without any help from others and without the use of documents or aids other than those stated. I have mentioned all used sources and cited them correctly according to established academic citation rules.

Göttingen, June 10th, 2026

Last in Translation? Evaluating Language-Invariant Reason Ranking in Numerical Fact-Checking

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1 Related Work

This section reviews the work most relevant to the thesis: LLM-based ranking, multilingual fact-checking, cross-lingual consistency, numerical reasoning, and preference-based fine-tuning. The central question is whether language invariance can be useful not only as something to measure, but also as something to optimize for: if a model is encouraged to behave consistently across languages, does its ranking and reasoning become more reliable?

This question connects directly to the three research questions of the thesis. RQ1 asks whether optimizing for language-invariant reasoning improves ranking performance. RQ2 examines how language invariance relates to reasoning quality across English, Spanish, and Arabic. RQ3 asks whether language-invariant training objectives lead to more robust multilingual reasoning models. The section first discusses LLMs as rankers and judges, including the presentation biases that affect them. It then reviews automated and multilingual fact-checking, before turning to cross-lingual consistency and the metrics used to measure it. Finally, it discusses multilingual representations, numerical reasoning across languages, and the preference-optimization and fine-tuning methods used in this thesis.

1.1 LLMs as Rankers and Rerankers

LLMs are now often used as rankers even without task-specific training, especially in retrieval and evaluation settings. One common approach is pairwise ranking prompting: the model compares two candidates at a time, and the final ranking is built by aggregating these pairwise preferences. This makes each decision easier for the model, but it is expensive because the number of comparisons grows quadratically with the number of candidates [30].

Listwise ranking takes a different approach. Instead of comparing candidates in pairs, the model sees all candidates at once and is asked to produce a complete ordering. This is more efficient at inference time, but it also makes the output depend on the full prompt and on the order in which the candidates are shown. Recent work has adapted learning-to-rank objectives to preference optimization in order to train listwise rankers more directly [24]. However, these generative ranking methods remain sensitive to presentation details such as candidate order and prompt format, rather than depending only on the content of the candidates [29].

This limitation motivates the independently scored design used in this thesis. Instead of asking the model to generate a full ranking directly, each candidate reasoning trace is scored separately and the final ranking is derived from those scores. This design is especially relevant for the numerical fact-checking task studied here, where the ranking should reflect the quality of the reasoning trace rather than artifacts of the prompt.

1.2 Biases in LLM-as-a-Judge Evaluation

Similar issues appear when LLMs are used as evaluators. Their judgments do not depend only on the quality of the content being assessed; they can also be shaped by how the options are presented. Studies of the LLM-as-a-judge setting show strong position bias, meaning that a candidate’s location in the prompt can affect whether it is preferred. Other effects, such as verbosity bias, have also been observed [41]. These biases matter for ranking because they can change which candidate is selected, even when the underlying content has not changed.

For this thesis, the most important issue is that LLM judges are often not language-invariant. In other words, the same model may give different judgments when the same evaluation is carried out in different languages. One

recent study examined five model families, five tasks, and twenty-five languages, and found that multilingual judge agreement was low, with an average Fleiss' κ of roughly 0.3. The study also found that multilingual pre-training and larger model scale did not reliably remove this inconsistency [12]. This kind of cross-lingual instability is exactly the problem that the research questions in this thesis address.

1.3 Automated and Multilingual Fact-Checking

Automated fact-checking is increasingly treated as a reasoning and assessment problem for LLMs. Rather than only asking whether a model can produce a label, recent work studies how the formulation of the task affects the reliability of model-based verification for real-world claims [35]. Numerical claims make this problem especially difficult. A small change in scope, units, dates, or quantitative interpretation can be enough to change the correct verdict.

The QuanTemp benchmark was introduced to study numerical claim verification in a real-world, open-domain setting, with claims paired with supporting evidence [45]. Later work builds on this by generating multiple reasoning traces for each claim and using a verifier to choose the traces that are most useful for reaching the correct verdict [4]. The CLEF CheckThat! 2026 Task 2 shared task follows this verifier-guided setup for multilingual numerical claim verification. In this task, systems work with English, Spanish, and Arabic claims, rank candidate reasoning traces, and then derive a claim-level verdict [46].

There is also a broader line of work on multilingual fact verification. X-Fact collects naturally occurring claims across twenty-five languages from eleven language families, together with expert veracity labels. Results on this benchmark show that even strong models achieve only modest accuracy, highlighting how difficult it is to transfer fact-checking ability across languages [13]. XFEVER extends the FEVER setting by translating claims and evidence into multiple languages, making it possible to study zero-shot and translate-train cross-lingual verification directly. It also finds that performance varies substantially by language and generally remains behind English [3]. Complementing these verification benchmarks, X-FACTR probes factual knowledge in multilingual models using native-speaker-authored fill-in-the-blank queries in twenty-three languages, showing that factual recall also differs across languages [18].

Together, these studies show why multilingual fact-checking is both important and difficult. They also provide the practical motivation for this thesis: if fact-checking systems behave differently across languages, then language invariance becomes a useful objective to study directly.

1.4 Language Invariance and Cross-Lingual Consistency

The main idea behind language invariance is simple: if two inputs express the same claim in different languages, a robust model should not change its judgment just because the language changed. This is especially important for the present task, where the model is expected to reason about the same numerical claim in English, Spanish, and Arabic. To make this idea useful experimentally, however, we need to be clear about what kind of consistency is being measured and how it is separated from accuracy.

Defining consistency. A useful starting point is the formal definition of consistency as stability under meaning-preserving changes to the input. In this view, a model is consistent if it gives non-contradictory predictions for two quasi-paraphrases of the same query. The consistency score is then the proportion of paraphrase pairs that lead to agreeing predictions [9]. This definition is important because it makes consistency separate from accuracy. A model can be consistent and still be wrong, or it can be accurate on average while giving inconsistent answers to equivalent inputs.

The same idea naturally extends from paraphrases within one language to translations across languages. If two versions of a claim have the same meaning, then a language-invariant model should give the same output for both. Prior work shows that this is not guaranteed. Multilingual factual-consistency studies find that encoder

models such as mBERT and XLM-R are no more consistent across languages than a monolingual model is across English paraphrases, and that inconsistency increases when typologically distant languages are included [10]. In this thesis, language invariance is therefore treated as cross-lingual consistency: agreement between a model’s outputs for translated versions of the same input, measured separately from whether those outputs are correct.

Measuring cross-lingual consistency. Because consistency and accuracy are different properties, cross-lingual consistency should be evaluated on its own. Work on factual knowledge consistency shows that multilingual models often have low cross-lingual consistency, that larger model size alone does not reliably solve the problem, and that consistency is strongly related to subword-vocabulary overlap between languages [28]. This suggests that scripts and tokenization can affect cross-lingual behavior, not only reasoning ability.

Different output types require different agreement measures. For categorical outputs such as claim verdicts, standard inter-rater agreement statistics are appropriate. Cohen’s κ measures agreement between two raters [5], Fleiss’ κ extends this idea to more than two raters [11], and Krippendorff’s α provides a more general reliability coefficient that can handle missing data [21]. Fleiss’ κ is also the statistic used in the multilingual LLM-as-a-judge work discussed earlier [12]. For ranked outputs, agreement can instead be measured with rank-correlation statistics such as Kendall’s τ and Spearman’s ρ [19, 42], or with top- k overlap measures such as Jaccard similarity.

One caveat is important. Agreement in behavior does not necessarily mean that the model uses the same internal representation across languages. Recent work shows that models can answer consistently across languages while still storing knowledge separately, especially for languages written in different scripts [16]. For this reason, the thesis treats invariance as an observable behavioral property, not as proof of a fully shared multilingual reasoning mechanism.

A measurement framing for this thesis. In this thesis, language invariance is measured operationally. For a claim c expressed in the language set $\mathcal{L} = \{\text{English, Spanish, Arabic}\}$, let $\hat{y}_\ell(c)$ be the verdict predicted from the language- ℓ version of the claim. Our primary metric is *verdict invariance*: whether the predicted claim verdict remains stable across languages. Concretely, we report the fraction of aligned claims for which all language-conditioned predictions agree, and we compute Fleiss’ κ over the verdicts $\{\hat{y}_\ell(c)\}_{\ell \in \mathcal{L}}$ by treating the predictions from the three languages as raters of the same claim [11]. We also report pairwise Cohen’s κ [5] and Krippendorff’s α [21] as supporting diagnostics.

The system also produces a ranking of candidate reasoning traces. For a claim c , let $r_\ell(c)$ denote the trace ordering produced from the language- ℓ version of the claim. As a secondary diagnostic, we measure ranking-level invariance by comparing these orderings across languages using mean pairwise Kendall’s τ , Spearman’s ρ [19, 42], and top- k Jaccard overlap over the ranked trace indices. These ranking metrics are not the primary invariance measure, but they help show whether cross-lingual stability appears in the trace ranking itself or only in the final verdict.

Finally, the training objective uses a different level of invariance. During training, invariance is encouraged at the score level: the model is penalized when it assigns different normalized score vectors to the same aligned traces across languages. This score-level regularizer is related to the consistency objectives discussed in Section 1.7, while the exact formulation is given in the methodology section. Throughout the thesis, invariance is reported separately from task accuracy. This separation is essential for RQ2, because a model may be consistent but wrong, or accurate but unstable across languages [9, 28].

1.5 Cross-Lingual Representations and the Latent Language

Language invariance is only plausible if multilingual models share at least some representation of meaning across languages. A large body of work suggests that this is partly the case. Studies of mBERT show that it can transfer across languages in a zero-shot setting, even across different scripts, although transfer quality depends

on factors such as typological similarity [27]. Other work shows that shared multilingual structure can emerge even without a shared vocabulary [52]. Large-scale multilingual pre-training, as in XLM-R, further improves transfer across many languages [6]. Explicit alignment methods can also bring contextual representations closer across languages and improve cross-lingual transfer [2]. A recent survey provides a broader overview of these alignment ideas and how they are evaluated [15].

For decoder-only LLMs, recent interpretability work points to a related but more complicated picture. Logit-lens analyses suggest that models such as Llama process multilingual inputs through intermediate representations that are often closest to English before producing an answer in the target language. This has been described as operating through a latent language [51]. Similarly, other work finds that semantic decisions in multilingual LLMs are made in a representation space close to English, and that interventions are most effective when applied in that space [37]. Representational similarity analyses also suggest that reasoning-like behavior is only partly aligned across languages [43].

These findings support one of the assumptions behind this thesis: if multilingual models already share part of their semantic space, then language-invariant training may encourage the model to use that shared space more consistently. In that sense, invariance can act as an inductive bias for more robust multilingual reasoning (RQ3). However, this should be interpreted carefully. A model can give the same answer in different languages without necessarily using the same internal representation in each case [16]. Therefore, the thesis treats behavioral invariance as useful evidence of cross-lingual stability, but not as proof of a fully language-independent reasoning mechanism.

1.6 Numerical Reasoning Across Languages and Value-Aware Representations

Numerical reasoning is a known weakness of LLMs. Even when the required operation is simple, models can behave unreliably. Probing studies describe this as a fragile number sense, where models fail on elementary numerical tasks that should in principle be straightforward [33]. Smaller diagnostic tests reach similar conclusions [32]. Benchmarks such as NumGLUE [26] and broader quantitative-reasoning evaluations [22] show that numerical reasoning is a demanding test of whether a model understands quantities, rather than only the surrounding text.

One reason this is difficult is that standard token representations do not encode numerical magnitude directly. A number is usually treated as a sequence of tokens, and the model has to infer its value from that surface form. Earlier probing work showed that some embeddings capture numerical magnitude surprisingly well, but also that subword tokenization limits exact numeracy [48]. Value-aware approaches address this problem by encoding the numerical value of a span, instead of relying only on its textual representation [8]. Related number-encoding methods expose canonical numerical information more directly to the model [38].

This issue becomes even more important in a multilingual setting. The task studied in this thesis is not only numerical, but also cross-lingual: the model must handle numerical claims in English, Spanish, and Arabic. Multilingual chain-of-thought evaluations, including the MGSM benchmark, show that quantitative reasoning quality varies across languages even when the underlying arithmetic is the same [40]. This means that language can affect numerical reasoning even when the mathematical problem itself has not changed.

Recent work makes this point even more directly by isolating the surface form of the number. When the same arithmetic problem is written with different numeral scripts or formats, model accuracy can drop substantially for underrepresented scripts. This effect is partly linked to tokenization, although targeted prompting and explicit numeral mapping can reduce the gap [34]. Related analyses show that models also struggle to infer the compositional structure of numerals across languages unless the relevant operations are made symbolically explicit [1].

These findings matter for this thesis because numeral script is another source of cross-lingual variation. If a model changes its quantitative judgment when the same number is written in a different script, then numerical

fact-checking becomes a particularly demanding setting for language invariance. Value-aware numerical representations and numerical normalization are therefore relevant not only as a way to improve numerical accuracy, but also as a possible way to reduce cross-lingual inconsistency. This is especially important for the Arabic part of the task, where script and numeral format may introduce additional variation.

1.7 Preference Optimization and Parameter-Efficient Fine-Tuning

The training setup in this thesis builds on two practical lines of work: preference optimization and parameter-efficient fine-tuning. Direct Preference Optimization (DPO) trains a model from pairwise preferences by increasing the relative likelihood of preferred outputs over rejected ones, while keeping the model close to a reference policy. This makes it possible to optimize preferences without training a separate reward model or using reinforcement learning [31].

For the fine-tuning itself, parameter-efficient methods are important because the models used in this thesis are too large to update fully under a modest GPU budget. Low-Rank Adaptation (LoRA) addresses this by freezing the base model and training only a small set of low-rank update matrices [14]. Quantized LoRA (QLoRA) reduces memory use further by loading the base model in low precision and training adapters on top of it [7]. These methods make it feasible to adapt open-weight instruction-tuned models such as Mistral-7B [17], Qwen3-8b[44] and Llama-3.1-8B-Instruct [25], which are the scale of models used in the experiments.

Recent work has also started to use these kinds of objectives for cross-lingual consistency. Consistency regularization penalizes a model when its predictions change under meaning-preserving perturbations, such as code-switching or machine translation [53]. Cross-lingual self-alignment extends the preference-optimization idea by sampling answers in several languages, selecting the most self-consistent answer as the preferred target, and applying DPO to align the model’s knowledge across languages [49]. Direct Consistency Optimization is even closer to the goal of this thesis: it adapts the DPO formulation to align completion likelihoods across parallel prompts in different languages, without requiring an explicit reward model [23].

These methods show that preference optimization can be used for consistency, not only for general output quality. However, most of this work evaluates consistency through factual question answering or broad benchmarks. It does not directly test whether optimizing for cross-lingual consistency improves ranking performance in a numerical fact-checking task. That is the gap addressed by RQ1: whether an invariance-oriented objective can improve the ranking itself, rather than simply making outputs agree across languages.

1.8 Positioning of This Thesis

The work reviewed above points to a clear gap. LLM rankers and judges are sensitive to presentation details such as candidate order and prompt format [29, 41]. They are also often not language-invariant: the same model can produce different verdicts or rankings for semantically equivalent inputs in different languages, and this inconsistency is not reliably solved by larger model scale or multilingual pre-training [10, 12, 28]. At the same time, multilingual models do appear to learn partly shared, partly English-centric representations [6, 51]. This suggests that language-invariant behavior may be attainable, while still leaving open the caution that behavioral agreement does not necessarily prove a shared internal representation [16].

The numerical side of the task adds another layer of difficulty. Prior work shows that numerical reasoning is fragile and only partly captured by standard token representations [33, 48]. It is also sensitive to numeral script and format across languages [34]. This matters for the present thesis because the task is not only multilingual, but also numerical: the model must rank reasoning traces for claims whose quantities may appear in different linguistic and script forms. Finally, recent work shows that preference optimization and consistency regularization can be aimed at cross-lingual consistency rather than output quality alone [23, 49, 53]. What remains unclear

is whether treating language invariance as a training objective actually improves multilingual reasoning in a trace-ranking fact-checking setting.

This thesis addresses that gap using the CLEF CheckThat! 2026 Task 2 setting, where systems rank candidate reasoning traces and produce a final verdict for numerical claims in English, Spanish, and Arabic. The three research questions separate the problem into ranking performance, the relationship between invariance and reasoning quality, and broader multilingual robustness.

RQ1: Does optimizing for language-invariant reasoning improve ranking performance? Prior work shows that consistency can be optimized directly [23, 49, 53], but most of this work evaluates the result on factual question answering or general benchmarks. RQ1 asks whether the same idea helps in a ranking-based fact-checking task. In particular, it tests whether adding a language-invariance objective to the trace scorer improves the official ranking metrics, rather than only making outputs more similar across languages. This connects to work on ranking bias [29, 30, 41]: if rankings are affected by irrelevant variation in the input, then reducing language-driven variation may make the ranking more reliable.

RQ2: What is the relationship between language invariance and reasoning quality across languages? Consistency is not the same as accuracy [9, 28]. A model can be stable across languages but wrong, or accurate in one language while unstable across others. RQ2 therefore measures invariance and per-language reasoning quality separately, then analyzes how they relate across English, Spanish, and Arabic. The thesis does not assume that the relationship is simple or monotonic. Prior work suggests that cross-lingual consistency can be affected by vocabulary overlap and script [28], and that agreement across languages does not necessarily imply shared internal representations [16]. For this reason, RQ2 treats the link between invariance and reasoning quality as an empirical question.

RQ3: Can language-invariant training objectives lead to more robust multilingual reasoning models? The representational evidence reviewed above suggests that multilingual models share part of their semantic space [6, 37, 51]. RQ3 asks whether a language-invariant objective can use this shared structure as an inductive bias for more robust multilingual reasoning. This is tested across English, Spanish, and Arabic, with particular attention to Arabic because script and tokenization can introduce additional variation [34]. The numerical dimension is also important: value-aware numerical representations and number-encoding methods are considered as complementary ways to reduce variation caused by the surface form of quantities [8, 38].

2 Background

This section introduces the concepts and terminology that the rest of the thesis assumes. It is written for a reader who is comfortable with general machine learning but not necessarily with large language models, fact-checking, ranking, multilingual evaluation, or the numerical and statistical tools we use. We begin with large language models and how they are adapted to downstream tasks, then describe text ranking and the basic classification machinery on which our system is built. We then introduce automated fact-checking and the specific numerical-claim-verification task studied here, followed by the multilingual notions of cross-lingual transfer and language invariance that motivate the thesis, a short treatment of numerical reasoning in language models, and finally the evaluation metrics and agreement statistics used throughout. Readers already familiar with a given topic can safely skip the corresponding subsection; the later sections cross-reference the definitions established here.

2.1 Large Language Models

A *large language model* (LLM) is a neural network trained to model the probability of natural language text. Modern LLMs are built on the *Transformer* architecture [47] and are typically *decoder-only*: they are trained by *next-token prediction*, that is, to predict the next unit of text given all the preceding units. Formally, for a sequence

of tokens $x_1, \dots, x_T$ the model factorizes the joint probability as $\prod_{t=1}^T p_\theta(x_t | x_{<t})$ and is optimized to maximize this likelihood over a very large corpus. After this *pre-training* stage the model has absorbed broad linguistic and world knowledge, which can then be steered toward specific tasks.

Tokens and tokenization. LLMs do not operate on characters or whole words but on *tokens*: sub-word units produced by a *tokenizer*. A tokenizer is learned on the pre-training corpus and splits text into frequent fragments, so that common words become single tokens while rare words are broken into several pieces. This *subword* scheme—most commonly byte-pair encoding [39]—keeps the vocabulary finite while still being able to represent any string. Tokenization is not a neutral detail for this thesis: the way a tokenizer splits numbers and non-Latin scripts directly affects how well a model can reason about quantities across languages, as discussed in Section 2.7.

Instruction tuning. A raw pre-trained model predicts plausible continuations but does not reliably follow instructions. *Instruction tuning* is a further training stage in which the model is fine-tuned on (instruction, response) examples so that it learns to carry out tasks phrased as natural-language requests. The resulting *instruction-tuned* models—for example Mistral-7B-Instruct [17], Llama-3.1-8B-Instruct [25], and Qwen3-8B [44], the three *backbones* used in this thesis—can be prompted to rank items, classify text, or answer questions without any task-specific training. The number in each name (7B, 8B) refers to the count of trainable parameters in billions, which is one rough proxy for model capacity.

2.2 Adapting Language Models to a Task

There are two broad ways to make an LLM perform a particular task: *prompt* it without changing its weights, or *fine-tune* some or all of its weights on task data. The thesis uses both, so we introduce each.

Prompting and in-context learning. In *zero-shot* prompting the model is simply given an instruction and the input, and asked to produce the output; in *few-shot* prompting the prompt additionally contains a handful of solved examples, from which the model infers the pattern—a behaviour known as *in-context learning*. Prompting requires no training and no gradient updates, which makes it a natural *baseline*, but its behaviour is sensitive to the exact wording and formatting of the prompt.

Chain-of-thought reasoning. A particularly important prompting technique for this thesis is *chain-of-thought* (CoT) prompting, in which the model is encouraged to produce a sequence of intermediate reasoning steps before committing to its final answer, rather than emitting the answer directly [50]. Making the reasoning explicit in this way substantially improves performance on tasks that require multiple steps, such as arithmetic and multi-hop questions, and it also makes the model’s reasoning inspectable. Chain-of-thought is directly relevant here because the “reasoning traces” that our task asks a system to rank (Section 2.5) are exactly such chains of thought: each trace is a step-by-step argument, generated by a language model, that works through a claim and its evidence and ends in a proposed verdict. The thesis does not generate these traces; it takes them as given and studies how to *select* and *rank* the most reliable ones.

Full fine-tuning versus parameter-efficient fine-tuning. *Fine-tuning* continues training the model on labelled task data. Updating all parameters of a 7–8B model (*full fine-tuning*) is expensive in memory and compute. *Parameter-efficient fine-tuning* (PEFT) instead freezes the pre-trained weights and trains only a small number of new parameters. The most common PEFT method is *Low-Rank Adaptation* (LoRA) [14]: for each adapted weight matrix, LoRA learns a pair of small low-rank matrices whose product is added to the frozen weight, so that only these compact *adapters* are trained. The number of trainable parameters is controlled by the adapter *rank* r , and the update is scaled by a factor α .

Quantization and QLoRA. *Quantization* stores model weights at lower numerical precision (for example 4 bits instead of 16 or 32) to reduce memory. *Quantized LoRA* (QLoRA) [7] combines the two ideas: the frozen base

model is loaded in 4-bit precision while the LoRA adapters are trained on top of it in higher precision. This makes it feasible to fine-tune models with billions of parameters on a modest GPU budget, which is the regime in which our experiments operate. Figure 1 illustrates how the low-rank adapters are added alongside the frozen weights.

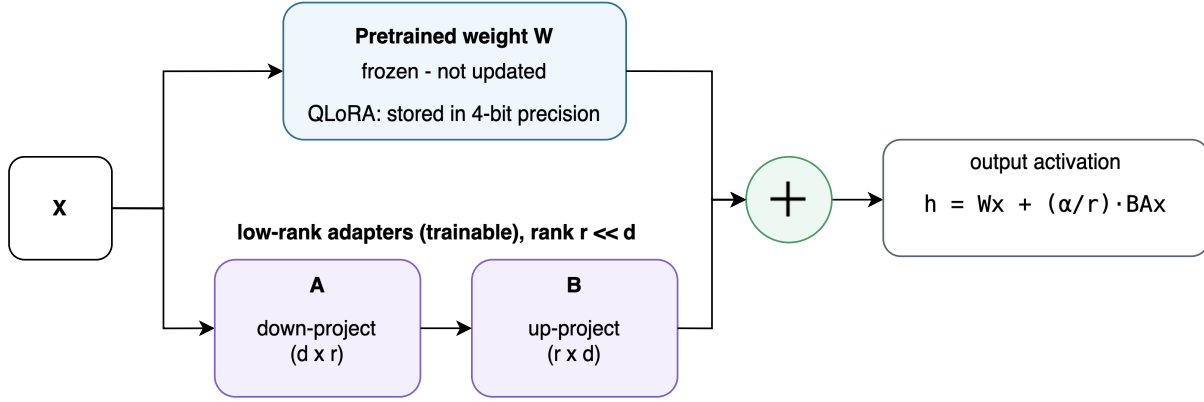


Fig. 1. LoRA and QLoRA. The pretrained weight matrix W is frozen; a small pair of low-rank matrices A and B (with rank $r \ll d$) is trained and their product is added to the frozen weight, so the layer computes $h = Wx + (\alpha/r) BAx$ and only A and B are learned. QLoRA additionally stores the frozen weight W in 4-bit precision to reduce memory.

Preference optimization. Beyond supervised fine-tuning, models can be trained from *preferences*: pairs of a preferred and a rejected output for the same input. *Reinforcement Learning from Human Feedback* (RLHF) was the original approach, but it is complex because it requires training a separate reward model and running reinforcement learning. *Direct Preference Optimization* (DPO) [31] achieves a similar effect more simply: it directly adjusts the model to raise the likelihood of preferred outputs relative to rejected ones, while staying close to a reference model, without an explicit reward model or reinforcement learning. The thesis uses DPO as one of its ranking baselines.

2.3 Text Ranking with Language Models

Ranking is the task of ordering a set of candidate items by how well they satisfy some criterion; the closely related task of *re-ranking* reorders an already-retrieved candidate set. LLMs can be used as rankers in three standard formulations, which differ in how many candidates the model sees at once; Figure 2 contrasts them.

- **Pointwise.** Each candidate is scored independently, and the ranking is obtained by sorting the scores. This is the formulation our trace scorer uses.
- **Pairwise.** The model compares two candidates at a time and states which is better; a global ordering is assembled from the accumulated pairwise preferences [30]. This keeps each decision simple but requires a number of comparisons that grows quadratically with the number of candidates.
- **Listwise.** All candidates are presented in a single prompt and the model emits a complete ordering. This is efficient at inference but couples the decision to the whole input, including the order in which candidates are listed [24].

LLM-as-a-judge and presentation bias. When an LLM is used to *evaluate* or compare candidates, the setting is often called *LLM-as-a-judge*. A recurring problem is that such judgments are influenced by factors unrelated to content quality. *Positional* (or *position*) bias is the tendency to prefer a candidate because of *where* it appears in the

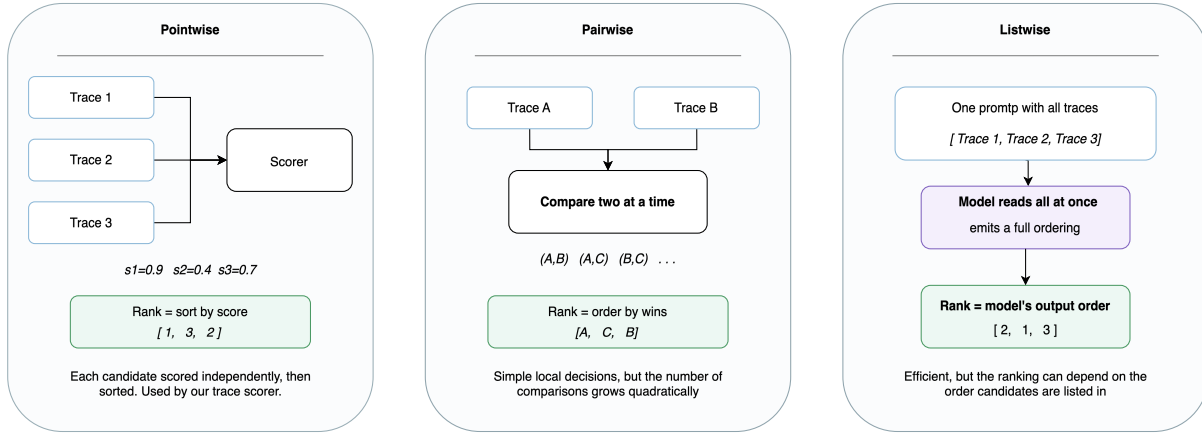


Fig. 2. The three standard formulations for ranking candidates with an LLM. *Pointwise* scoring evaluates each candidate independently and sorts by score (the formulation used by our trace scorer); *pairwise* ranking compares two candidates at a time and orders them by accumulated wins; *listwise* ranking presents all candidates in one prompt and reads off the model’s output order.

prompt rather than *what* it says; related effects include verbosity bias and sensitivity to prompt format [29, 41]. These biases matter for ranking because they can determine which candidate is chosen independently of its actual reliability, and they are one reason the thesis moves from generative ranking toward independently scored candidates.

2.4 Classification and Training Objectives

Our system reformulates ranking as *binary classification*, so we recall the relevant basics. A classifier maps an input to one or more real-valued *logits*. For *binary* classification one can use a single logit ℓ passed through the *sigmoid* function $\sigma(\ell) = 1/(1 + e^{-\ell})$ to produce a probability, or two logits (ℓ^{neg} , ℓ^{pos}) passed through the *softmax* function, which normalizes a vector of logits into a probability distribution. The model is trained by minimizing a *loss* that penalizes disagreement between the predicted probability and the true label; for classification this is the *cross-entropy* loss (its two-class special case is *binary cross-entropy*). The higher the probability the model assigns to the positive class, the higher we can consider its *score* for that input, and this score is exactly what we later sort on to obtain a ranking. A *classification head* is the small final layer that turns the model’s internal representation of an input into these logits.

2.5 Automated Fact-Checking and Numerical Claim Verification

Automated fact-checking is the task of assessing whether a *claim*—a statement that can be true or false—is supported by evidence. The step that assigns a truth value to a claim given evidence is called *claim verification* (or veracity prediction), and it is the part of the fact-checking pipeline this thesis is concerned with. It is worth distinguishing it from the earlier stages of a full fact-checking system, which typically also include *claim detection* (deciding which sentences are checkworthy claims) and *evidence retrieval* (finding documents relevant to a claim); we take claim detection as already done and focus on verification given retrieved evidence [35].

Core components: claims, evidence, and verdicts. A verification example has three core components. The *claim* is the statement to be checked. The *evidence* is a set of retrieved text snippets—the output of a *retrieval* step that searches a document collection for material relevant to the claim—against which the claim is assessed; in this

thesis the evidence is provided with each example, so we do not perform retrieval ourselves. The *verdict* (also called a veracity label) is the categorical outcome describing the claim’s relationship to the evidence, drawn here from the label space True, False, and Conflicting. In our specific task a fourth component is added on top of these: a set of candidate *reasoning traces* (chains of thought, in the sense of Section 2.2), each carrying its own proposed verdict, which the system must rank. *Numerical claims* are claims whose truth hinges on quantities—counts, dates, percentages, rankings, or comparisons—and they are especially demanding, because a small change in a number, a unit, or the scope of a comparison can flip the verdict [45].

Reasoning traces and verifier-guided verification. A recent approach to hard verification problems is *test-time scaling*: instead of producing a single answer, the model generates several independent *reasoning traces*—step-by-step chains of reasoning, each ending in a proposed verdict—and a separate component decides which traces to trust [4]. The component that selects or ranks traces is a *verifier*, and the overall setup is *verifier-guided*. The task studied in this thesis follows exactly this shape: for each claim it provides evidence and a set of candidate reasoning traces with their per-trace verdicts, and a system must (i) *rank* the traces by how useful they are for reaching the correct answer and (ii) derive a final *claim-level verdict*, typically from the highest-ranked traces. Because the traces are given and only their ranking and aggregation are learned, the task is best seen as a ranking-and-aggregation problem rather than free-form reasoning. Figure 3 shows the full pipeline and marks which parts are provided by the task and which are learned in this thesis.

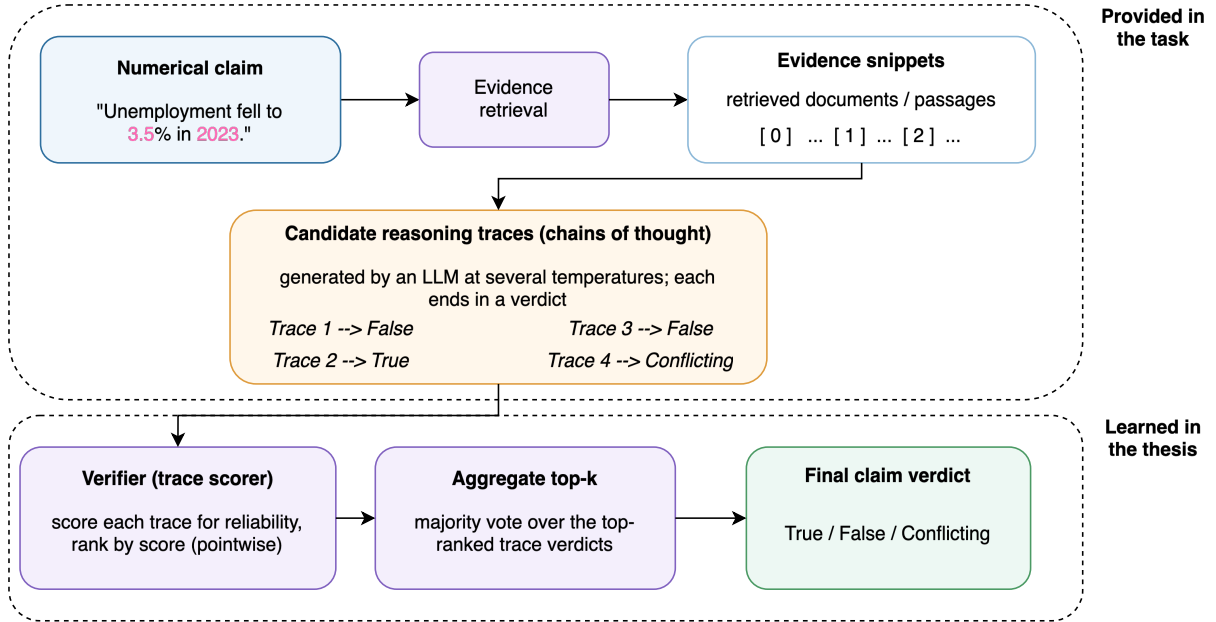


Fig. 3. Numerical claim verification framed as verifier-guided trace ranking. A numerical claim and its retrieved evidence, together with a set of candidate reasoning traces (each ending in a proposed verdict), are provided in the task. The system this thesis develops scores and ranks the traces, aggregates the top-ranked verdicts by majority vote, and outputs a final claim verdict from {True, False, Conflicting}.

Benchmarks and the shared task. QuanTemp is a benchmark of real-world numerical claims with associated evidence, introduced to study this problem in an open-domain setting [45]. The CLEF CheckThat! 2026 Task 2

shared task [46] adopts the verifier-guided formulation for *multilingual* numerical claim verification across English, Spanish, and Arabic, and is the setting in which our system [20] was developed. More broadly, multilingual fact-checking has its own line of benchmarks—for example X-Fact [13] and XFEVER [3]—which consistently report that transferring verification across languages is difficult and that performance varies widely by language.

2.6 Multilinguality and Cross-Lingual Consistency

A *multilingual* model is one trained on text in many languages; the practical hope is *cross-lingual transfer*, where ability learned in one language carries over to others. A large body of work shows that multilingual models learn substantially *shared* internal representations, so that semantically equivalent inputs in different languages are encoded similarly. For decoder-only LLMs, interpretability studies suggest that models process non-English input through intermediate representations that are close to English before mapping back to the input language, an effect described as operating in a *latent language* [37, 51]. This shared space is what makes it plausible, in principle, for a model to behave the same way regardless of the language of the input.

Consistency and language invariance. In practice, however, models are frequently *not* consistent. *Consistency* is defined as invariance of a model’s predictions under meaning-preserving changes to the input: if two inputs mean the same thing, a consistent model produces non-contradictory outputs [9]. When the meaning-preserving change is a *translation*, consistency becomes *cross-lingual consistency*, or *language invariance*: agreement of a model’s outputs across translations of the same input [10, 28]. This is the central property the thesis studies. A crucial subtlety, which the thesis takes care to respect, is that **consistency is decoupled from accuracy**: a model can be consistently wrong (agreeing across languages on an incorrect answer) or accurate but inconsistent (right on average yet disagreeing with itself across languages) [9, 28]. A further caveat is that answering the same way across languages does not by itself prove the knowledge is stored in a shared representation [16]. Because of this decoupling, invariance and accuracy must be *measured separately* before their relationship can be studied, which directly shapes the evaluation protocol of this thesis.

2.7 Numerical Reasoning in Language Models

Numeracy—the ability to reason about numbers—is a known weak point of language models. Probing studies report that models have a fragile “number sense” and fail on elementary numerical operations [33, 48]. Part of the difficulty is representational: the *magnitude* of a number (how big it is) is not explicitly encoded by standard token embeddings, and subword tokenization can split a numeral into pieces that obscure its value [48]. Two lines of remedy are relevant here. *Value-aware* representations encode the numeric *value* of a span rather than relying only on its surface tokens [8, 38]. Separately, the numeral *script* and *format* themselves matter across languages: the same quantity written in Arabic-Indic digits or with different decimal and thousands separators tokenizes differently, and accuracy can drop for under-represented scripts even when the underlying arithmetic is unchanged [1, 34]. This is why numeral script is an independent axis of cross-lingual variation, and why the thesis pays particular attention to it for Arabic.

2.8 Evaluation Metrics and Agreement Statistics

Finally, we define the metrics used to report results. They fall into three groups: classification quality, ranking quality, and cross-lingual agreement.

Classification metrics. For a class of interest, *precision* is the fraction of items predicted to be in the class that truly are, and *recall* is the fraction of true class members that were predicted. The *F1 score* is their harmonic mean, $F1 = 2PR/(P + R)$, which balances the two. *Macro-F1* averages the F1 score computed separately for each class,

giving every class equal weight regardless of how frequent it is; we use it for the claim-level verdict, which has three classes.

Ranking metrics. For a ranked list and a set of *relevant* items, $Recall@k$ is the fraction of relevant items that appear in the top k positions, and $Precision@k$ is the fraction of the top k positions occupied by relevant items. *Mean Reciprocal Rank at k* (MRR@ k) rewards placing a relevant item early: it is the reciprocal of the rank of the first relevant item (or zero if none appears in the top k), averaged over queries. In our task, a trace is “relevant” when its verdict matches the gold claim label, and $Recall@5$ is the primary ranking metric.

Inter-rater agreement. To quantify how much two or more sets of predictions agree—for example, the verdicts a model produces for the same claim in different languages—we borrow *inter-rater agreement* statistics from the study of human annotators, treating each language as a “rater.” The simplest measure is *raw agreement*, the fraction of items on which the raters give the same label, but this rewards agreement that would occur by chance. Chance-corrected coefficients address this: *Cohen’s κ* measures agreement between exactly two raters beyond chance [5]; *Fleiss’ κ* generalizes this to more than two raters [11]; and *Krippendorff’s α* is a reliability coefficient that also tolerates missing data [21]. All three equal 1 for perfect agreement and around 0 for chance-level agreement.

Rank-agreement. When the outputs being compared are *rankings* rather than labels, we use rank-correlation statistics. *Kendall’s τ* compares two orderings by the balance of *concordant* and *discordant* pairs (pairs ordered the same versus oppositely) [19], and *Spearman’s ρ* measures the correlation between the two sets of ranks [42]. Both range from -1 (reversed) through 0 (unrelated) to $+1$ (identical order). To focus on the top of the ranking, we also use the *Jaccard* overlap of the two top- k sets, defined as the size of their intersection divided by the size of their union. These rank-agreement measures let us ask not only whether a model reaches the same verdict across languages, but whether it ranks the underlying reasoning traces in the same way.

3 Methodology

This section explains how we investigate whether language invariance can serve as a useful training objective for multilingual reasoning. Our starting point is the supervised trace-ranking system developed for our CLEF Check-That! 2026 Task 2 submission [20, 46], which already includes value-aware numerical components—numeric embeddings and plain-text numerical normalization—aimed at reducing variation caused by differences in numeral script and format, particularly in Arabic. We use this system as the experimental backbone and extend it in two main ways.

First, we introduce a score-level language-invariance objective that encourages the model to assign similar scores to corresponding reasoning traces across languages. This allows us to examine whether optimizing for invariance improves ranking performance and robustness (RQ1 and RQ3). Second, we define an evaluation protocol that measures language invariance separately from task performance. This makes it possible to study the relationship between cross-lingual consistency and reasoning quality rather than assuming that more consistent predictions are necessarily more accurate (RQ2). Within this setting we also revisit the inherited value-aware numerical components as a complementary lever for the numerical, script-sensitive dimension of the task, now studied in combination with the invariance objective (RQ3).

The remainder of the section presents the task and dataset, followed by the trace-scoring architecture and its training objective. We then describe the language-invariance objective and the numerical components, introduce the baselines, and define the evaluation protocol. Finally, we provide the experimental configuration and the experiment matrix used for Mistral-7B, Llama-3.1-8B, and Qwen3-8B.

3.1 Task Formulation

We adopt the verifier-guided reranking formulation of CLEF CheckThat! 2026 Task 2 [46], which itself follows the test-time scaling setup for numerical claim verification [4, 45]. Each example consists of a claim c , a set of retrieved evidence snippets $E(c)$, and a set of candidate reasoning traces $R(c) = \{1, \dots, n_c\}$ that were generated by prompting a language model at several decoding temperatures to induce diverse reasoning paths. Each trace $i \in R(c)$ carries a trace-level verdict v_i , and the claim carries a gold claim-level label $y_c^\star$. The label space is $\mathcal{Y} = \{\text{True}, \text{False}, \text{Conflicting}\}$. A system must produce two outputs for every claim: a ranking $r(c)$ of the candidate traces by their usefulness for reaching the correct verdict, and a predicted claim verdict $\hat{y}_c \in \mathcal{Y}$.

The official evaluation is performed independently for each language $\ell \in \mathcal{L} = \{\text{English}, \text{Spanish}, \text{Arabic}\}$. Ranking quality is measured by Recall@5 over the traces whose verdict matches the gold claim label,

$$\text{Recall@5}(c) = \frac{|\{i \in r_{1:5}(c) : v_i = y_c^\star\}|}{|\{i \in R(c) : v_i = y_c^\star\}|}, \quad (1)$$

where $r_{1:5}(c)$ denotes the five top-ranked traces; claim-verdict quality is measured by macro-F1 over $\mathcal{Y}$; and the per-language score is the mean of macro-F1 and Recall@5. We additionally report MRR@5 as a secondary ranking diagnostic. A structural property of Equation (1) matters throughout: for a claim with m gold-matching traces, even a perfect ranking achieves at most $\min(5, m)/m$, and a claim with $m = 0$ contributes Recall@5 of zero. On the validation data this caps the attainable Recall@5 at 0.39, 0.32, and 0.28 for English, Spanish, and Arabic respectively, which bounds the headroom of every ranking method and must be kept in mind when reading results.

Generative ranking formulations—listwise and pairwise prompting, and DPO over complete ranking outputs—are sensitive to candidate order and prompt format rather than content alone [30, 31, 41], in line with the presentation-bias literature surveyed in section 1 [12, 29]. Supervised trace scoring, by contrast, assigns each claim-trace pair an independent relevance score and ranks deterministically by that score, removing the dependence on output order. We therefore take the trace scorer as the substrate on which the language-invariance objective is defined, because it exposes a continuous per-trace score that can be aligned across languages without having to reconcile discrete generated rankings.

3.2 Data

We use the multilingual numerical claim-verification dataset released for the shared task. The English portion is based on QuanTemp [45], while the Spanish and Arabic portions contain 2,808 and 3,260 claims, respectively, and come from the previous edition of the lab together with their evidence collections [46]. Each example contains the claim text, a collection of evidence snippets, a list of candidate reasoning traces (Reasoning_traces), the verdict produced by each trace (Verdict_list), and the gold claim label (label). We map both the gold labels and trace verdicts to the common label space introduced in Section 3.1. A trace is then marked as relevant when its normalized verdict matches the normalized gold label.

Complete and positive-only splits. We use two versions of each training split. The *complete* version contains every claim, including claims for which none of the candidate traces produces the correct verdict. The *positive-only* version excludes these claims. This distinction is necessary because a claim with no gold-matching trace provides no positive example for the binary trace scorer: every candidate trace is labelled as irrelevant. Including many such claims increases the imbalance between positive and negative traces without helping the model learn to distinguish useful reasoning from incorrect reasoning. We therefore apply this filtering only during training. All evaluation is performed on the complete splits so that the results reflect the full task, including claims for which Recall@5 is necessarily zero.

Aligned multilingual data for invariance. Training with the language-invariance objective requires parallel examples across languages. In particular, we need the same claim and candidate reasoning traces in different languages so that the scores assigned to corresponding traces can be compared directly. We construct these aligned data through machine translation, creating language-specific versions of each claim, its evidence, and its reasoning traces. During translation, we preserve the original dataset_index and trace ordering. Consequently, trace i for a given claim in one language is intended to correspond to trace i for the same claim in the other languages.

3.3 Trace-Scoring Architecture

Backbone models. We implement the trace scorer using three open-weight, instruction-tuned decoder models: Mistral-7B-Instruct-v0.3 [17], Llama-3.1-8B-Instruct [25], and Qwen3-8B [44]. Each backbone is loaded as a sequence-classification model. The pretrained transformer processes the claim-trace input, and a lightweight linear head maps its final representation to a relevance score.

Using three model families allows us to test whether the effects of the language-invariance objective generalize beyond a single backbone. This is particularly relevant to RQ2 and RQ3 because the models differ in their pretraining data and tokenizers, and previous work suggests that cross-lingual consistency is related to subword-vocabulary overlap [28, 34].

Scoring head. Given a claim-trace input x_i , the model produces a scalar score s_i representing the estimated relevance of trace i . We consider two configurations for the classification head. The default cross-entropy head produces two logits, $(\ell_i^{\text{neg}}, \ell_i^{\text{pos}})$. The corresponding probability that the trace is relevant is

$$p_i = \text{softmax}(\ell_i^{\text{neg}}, \ell_i^{\text{pos}})_{\text{pos}}.$$

For ranking, we use the logit margin

$$s_i = \ell_i^{\text{pos}} - \ell_i^{\text{neg}},$$

which produces the same ordering as the positive-class probability. The alternative binary cross-entropy head produces a single logit ℓ_i , which is used directly as the ranking score, $s_i = \ell_i$, with $p_i = \sigma(\ell_i)$.

Both configurations treat trace ranking as independent binary classification: each trace is scored without reference to the position or presentation order of the other candidates. The resulting continuous scores can then be used both to rank traces within a claim and to align corresponding traces across languages.

Parameter-efficient fine-tuning. We fine-tune each backbone using QLoRA [7, 14]. The base model is loaded in 4-bit NF4 precision with a bf16 computation type, and its original parameters remain frozen. Trainable low-rank adapters are added to all linear projections using target_modules=all-linear. Unless stated otherwise, the adapters use rank $r = 16$, scaling factor $\alpha = 32$, dropout 0.05, and no adapter bias.

The classification head is kept outside the quantized model and trained in full precision using modules_to_save to ["score"]. This is necessary because the head is initialized for the classification task and must learn the scale of the relevance scores. We also use PyTorch scaled dot-product attention and gradient checkpointing to reduce the memory required by long claim-evidence-trace inputs.

This configuration makes it possible to adapt 7–8B parameter models within a modest multi-GPU budget. It also leaves most of the pretrained multilingual representations unchanged, allowing us to investigate whether the shared latent representations identified in previous work [37, 51] can support language-invariant trace scoring.

Input construction. Figures 4 and 5 summarize the complete scoring process. Each claim and candidate trace is converted into a single input with the following structure:

```
Claim: {claim}
Evidence snippets:
```

```

[0] {evidence snippet}
[1] {evidence snippet}
...
Verdict: {trace verdict}
Justification: {cleaned trace justification}

```

Before constructing the input, we clean the trace justification by removing explicit “Label:” and “Justification:” markers, repeated verdict tokens, and unnecessary whitespace. The trace verdict remains available in its dedicated field, but removing repeated label information from the justification prevents it from dominating the reasoning text.

Input length is controlled at both the character and token levels. We limit the lengths of the claim, individual evidence snippets, and trace justification, as well as the number of evidence snippets included. The resulting sequence is then truncated to the model’s configured maximum length. Our compact configuration uses two evidence snippets and a maximum sequence length of 1,024 tokens, making it suitable for rapid experimentation. The long-context configuration includes up to twenty evidence snippets, allows up to 3,078 characters per snippet, and supports sequences of up to 15,000 tokens. Traces whose cleaned justifications contain fewer than three words are excluded because they provide too little reasoning content to be useful for training.

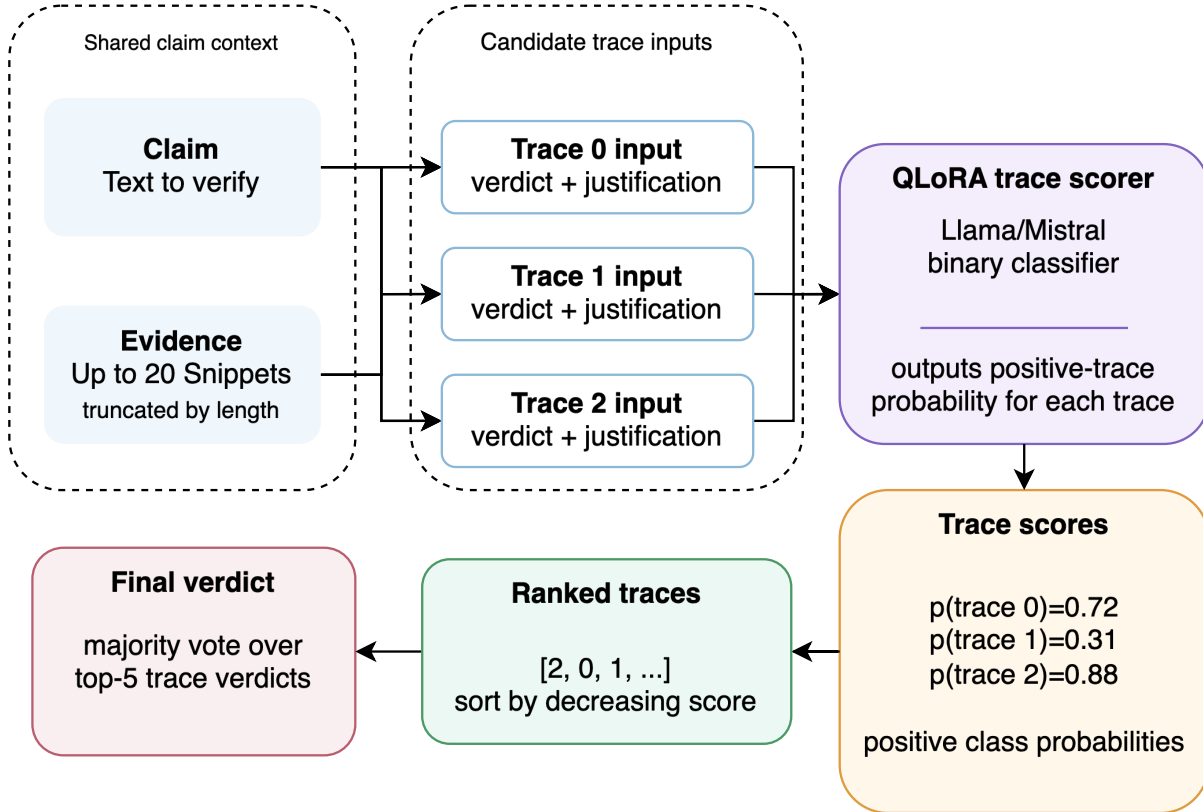


Fig. 4. Input-output flow of the supervised trace scorer. Candidate traces are scored independently, sorted by positive-class probability, and aggregated through the top-ranked trace verdicts.

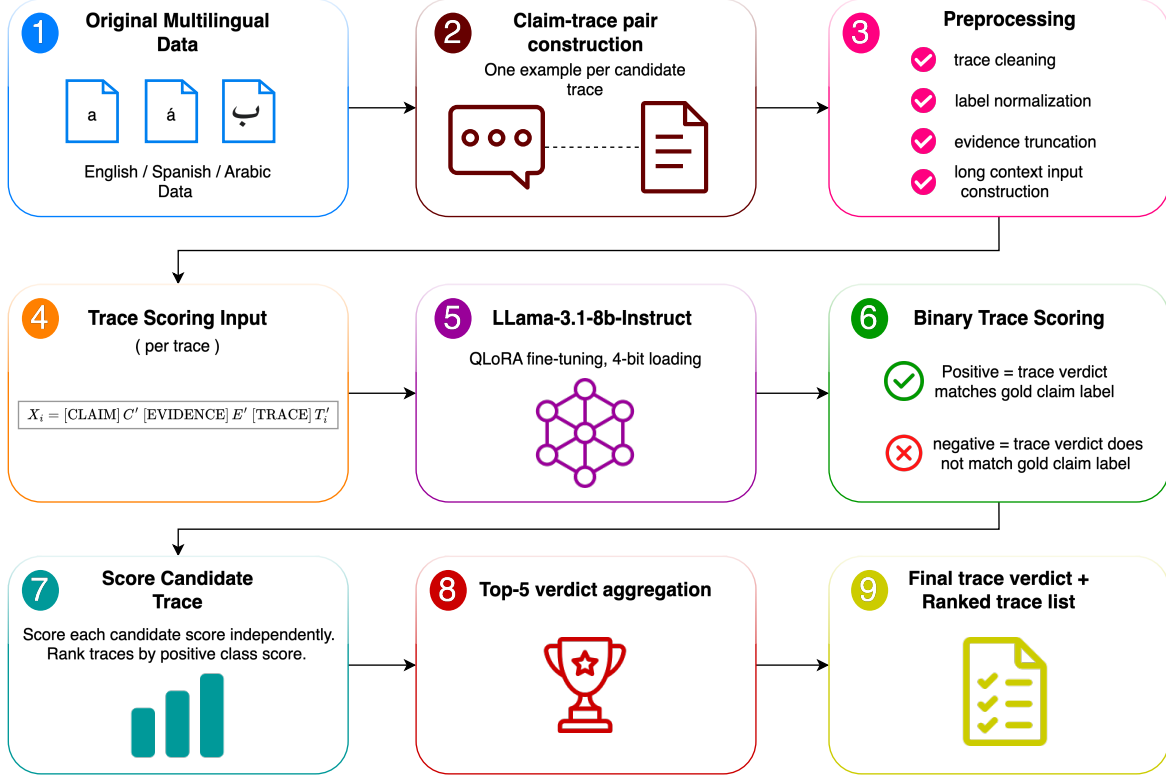


Fig. 5. Input-output flow of the supervised trace scorer. Candidate traces are scored independently, sorted by positive-class probability, and aggregated through the top-ranked trace verdicts.

3.4 Trace-Classification Objective

The trace scorer is trained as a binary classifier. For trace i belonging to claim c , the target label is defined as

$$z_i = \begin{cases} 1, & \text{if } v_i = y_c^*, \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

A trace is therefore considered relevant when its verdict agrees with the gold claim label. With the two-logit head, we minimize the standard cross-entropy loss between $(\ell_i^{\text{neg}}, \ell_i^{\text{pos}})$ and z_i . With the single-logit head, we use binary cross-entropy between $\sigma(\ell_i)$ and z_i . We refer to this classification term as the ranking loss, $\mathcal{L}_{\text{rank}}$, because it trains the scores from which the final ranking is constructed.

At inference time, every candidate trace is scored independently. The ranking is obtained by sorting the traces in decreasing order of score:

$$r(c) = \text{argsort}_{i \in R(c)}(-s_i).$$

When two traces receive the same score, their original trace indices are used to break the tie deterministically.

The claim-level verdict is derived from the ranked traces rather than predicted by a separate classification head. Specifically, we take a majority vote over the verdicts of the top- k traces, using $k = 5$ by default:

$$\hat{y}_c = \text{majority}(\{v_i \mid i \in r_{1:k}(c)\}). \quad (3)$$

If the vote is tied, the label that appears first in the ranking is selected. The ranking and claim-level verdict are therefore both determined by the same learned trace scores. Improving these scores can consequently affect both Recall@5 and macro-F1, even though the final verdict is produced through a separate majority-vote readout.

3.5 Language-Invariance Objective

The central methodological contribution of the thesis is to add, on top of $\mathcal{L}_{\text{rank}}$, a score-level regularizer that penalizes a model when it assigns different relevance profiles to the *same* traces across languages. This operationalizes the measurement framing of Chapter 1: invariance is encouraged at the level of continuous trace scores rather than discrete verdicts, in the spirit of cross-lingual consistency objectives [23, 49, 53], but—unlike that prior work—it is tied directly to a ranking task so that its effect on ranking quality can be measured (RQ1). Figure 6 gives an overview of the objective.

Aligned multilingual batching. Training consumes the aligned data of Section 3.2. We first build trace-scorer features independently per language, tagging each with its claim identity (dataset_index), trace index, and a language id. We then form *aligned claim groups*: for each claim that is present in every language, we take the set of trace indices that survive preprocessing in all languages, and emit one group containing, for that claim, every (language, common-trace) example. Groups are the unit of batching, so that whenever the invariance term is computed the batch is guaranteed to contain the parallel scores it needs; a claim that has no common trace across languages contributes no invariance signal and is dropped from the aligned set (it can still contribute to $\mathcal{L}_{\text{rank}}$). The standard classification labels and metadata are carried through the collator unchanged, so $\mathcal{L}_{\text{rank}}$ is identical to the monolingual case.

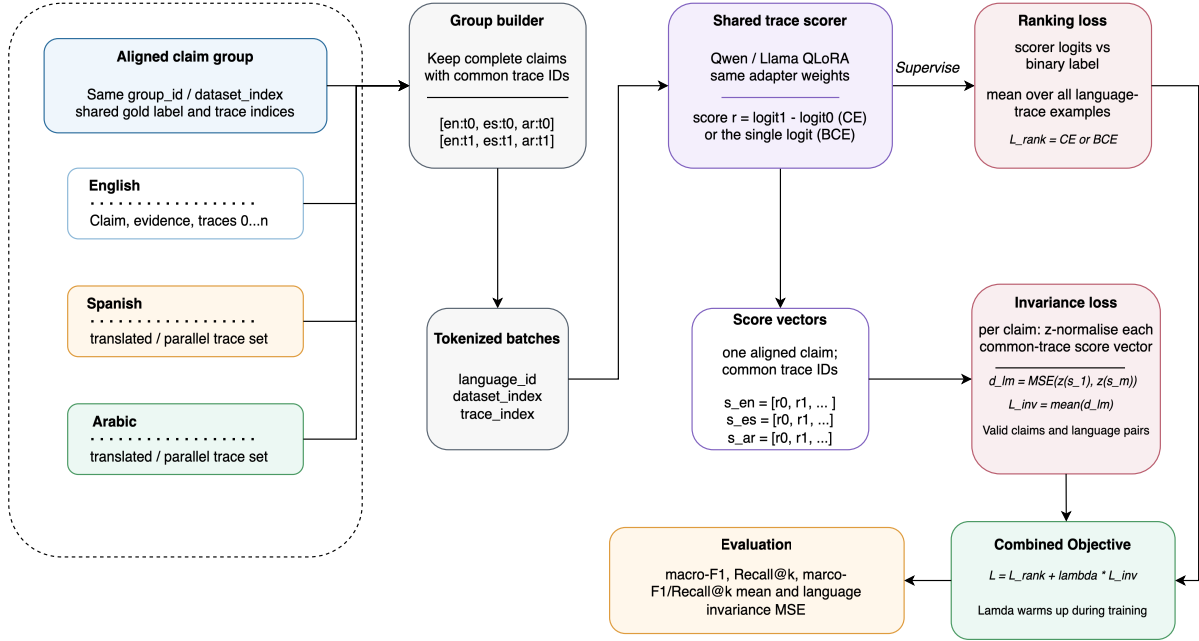


Fig. 6. Language-invariant multilingual trace-scoring pipeline. Parallel English, Spanish, and Arabic claim–trace examples are aligned by claim and trace indices and processed by a shared QLoRA scorer. Training combines binary trace-ranking loss with pairwise language-invariance loss computed over normalized, aligned score vectors.

Score normalization. Absolute score scales are not comparable across languages—a model may be systematically more or less confident in one language even when its *relative* ordering of traces is identical. Because ranking depends only on the relative scores within a claim, we compare languages after removing per-claim, per-language location and scale. For a claim c , language ℓ , and the set of common trace indices T_c , let $\mathbf{s}_c^{(\ell)} = (s_{c,i}^{(\ell)})_{i \in T_c}$ be the score vector; we standardize it as

$$\tilde{\mathbf{s}}_c^{(\ell)} = \frac{\mathbf{s}_c^{(\ell)} - \text{mean}(\mathbf{s}_c^{(\ell)})}{\text{std}(\mathbf{s}_c^{(\ell)}) + \varepsilon}, \quad (4)$$

with a small ε for numerical stability. Standardizing makes the objective penalize differences in the *shape* of the score profile—which trace is ranked above which, and by how much relative to the spread—rather than harmless constant offsets, aligning the regularizer with what the ranking metric actually rewards.

Invariance loss. For each aligned claim group with at least two languages, we penalize the mean squared difference between the standardized score vectors of every language pair, and average over claims:

$$\mathcal{L}_{\text{inv}} = \text{mean}_c \frac{1}{\binom{|\mathcal{L}_c|}{2}} \sum_{\ell < \ell'} \|\tilde{\mathbf{s}}_c^{(\ell)} - \tilde{\mathbf{s}}_c^{(\ell')}\|_2^2 / |T_c|, \quad (5)$$

where $\mathcal{L}_c$ is the set of languages available for claim c . The total training objective is

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \lambda_{\text{inv}}(t) \mathcal{L}_{\text{inv}}, \quad (6)$$

where the weight $\lambda_{\text{inv}}(t)$ is warmed up linearly from 0 to its target value over the first fraction ρ of training steps and held constant thereafter,

$$\lambda_{\text{inv}}(t) = \begin{cases} \lambda^* \cdot \frac{t}{\rho T}, & t < \rho T, \\ \lambda^*, & t \geq \rho T, \end{cases} \quad (7)$$

with T the total number of optimization steps. The warm-up lets the head first learn a sensible score scale from $\mathcal{L}_{\text{rank}}$ alone, before invariance pressure is applied; applying it from step zero risks collapsing the scores toward a trivially consistent but uninformative solution, the cross-lingual analogue of being “consistently wrong” [9, 28]. Setting $\lambda^* = 0$ recovers the plain multilingual trace scorer, which gives a clean ablation: the same data, batching, and architecture, with and without the invariance term. We log $\mathcal{L}_{\text{rank}}$, $\mathcal{L}_{\text{inv}}$, and $\lambda_{\text{inv}}(t)$ separately throughout training so that the two pressures can be monitored independently.

3.6 Value-Aware Numerical Components

Numerical expressions can vary considerably across languages even when they represent the same quantity. For example, a value may be written using Arabic-Indic or Persian digits instead of Western Arabic digits, or it may use different conventions for decimal and thousands separators. These surface differences can produce very different tokenizations and are a known source of cross-lingual inconsistency [1, 34, 48]. To reduce this variation, we introduce two optional value-aware components. Both can be used with the standard trace scorer and combined with the language-invariance objective.

Numeric canonicalization. Both components use the same numeric canonicalization procedure. Numeric spans are first identified using a regular expression that supports Western Arabic, Arabic-Indic, and Persian digits, as well as percentage signs, per-mille signs, and numbers containing comma or decimal separators. Each detected expression is converted to ASCII digits, and its separators are interpreted and rewritten using a consistent decimal format. Percentage and per-mille expressions are converted to their corresponding decimal values. The result is then parsed as a numeric value and rendered as a canonical decimal string.

This process maps different surface forms of the same quantity to a common representation. For example, quantities written using different digit scripts or separator conventions receive the same canonical value. The two numerical components described below use this representation in different ways.

Value-aware numeric embeddings. Figure 7 illustrates the value-aware embedding pipeline. Each detected number is preceded by a special `<num>` token while its original surface form remains in the input. The corresponding canonical value is passed to a small learned value encoder. This encoder represents the canonical string character by character using a fixed vocabulary containing digits, a sign, a decimal point, and an exponent marker. The character embeddings are processed by a single-layer GRU and projected to the hidden size of the backbone model.

At every `<num>` position, the standard token embedding is replaced by the resulting value vector. The transformer therefore receives an explicit representation of the number’s value while still retaining its original textual form. This reduces its dependence on the subword tokenization of the numeral and follows previous work on value-aware numerical representations [8, 38].

To support this component, the `<num>` token is added to the tokenizer and the model’s input embedding matrix is resized. The value encoder is trained jointly with the LoRA adapters and classification head. It is also saved with the trained adapter so that the same numerical representation can be restored during inference.

Plain-text numeric normalization. The second component provides a simpler way of exposing canonical values to the model. Rather than changing the embedding layer, it appends a short plain-text section headed `Normalized numbers:` to the trace-scorer input. This section lists the numbers found in the claim, evidence, and reasoning trace. When a number’s original and canonical forms differ, it is shown as a `surface=canonical` pair.

This approach requires no change to the model architecture because the normalized values are presented as ordinary input text. It provides a lightweight alternative to the learned value encoder and helps determine whether any improvement comes simply from making canonical values visible to the model or from the learned embedding mechanism itself. The normalization procedure also ignores small leading integers used as list markers, such as “1.” or “2)”, so that they are not incorrectly treated as quantities.

3.7 Baselines

We compare the trace scorers against the progression of baselines established for the shared task, which also serve as the reference points for the invariance analysis.

- **Prompt baseline.** A frozen instruction-tuned backbone, used without any task-specific fine-tuning, is our primary untrained reference. Given the claim, evidence, and candidate traces, it is prompted to produce a trace ranking and a claim verdict directly. We run this prompted baseline in two modes, listwise and pairwise, described next.
- **Listwise prompting** (Figure 8). The model receives the claim, evidence, and all candidate traces in one prompt and is asked to emit strict JSON containing a permutation of trace indices and one verdict. Despite an explicit instruction not to copy the input order, the produced ranking remains sensitive to the original trace order, exhibiting the positional bias documented for listwise rankers [29, 41].
- **Pairwise prompting** (Figure 9). Ranking is reformulated as a set of two-trace comparisons whose accumulated wins define the ordering [30]. To counter A/B position bias we use a *balanced* orientation, which alternates which trace is shown first, and a *bidirectional* orientation, which evaluates each pair in both orders and counts a win only when both orders agree. These reduce but do not remove the order effect, at a cost of $n(n-1)/2$ or $n(n-1)$ model calls per claim.
- **Direct Preference Optimization (DPO)** (Figures 11 and 10). An English LoRA model, initialized from a supervised fine-tuned adapter on Mistral-7B, is trained with DPO [31] on automatically constructed

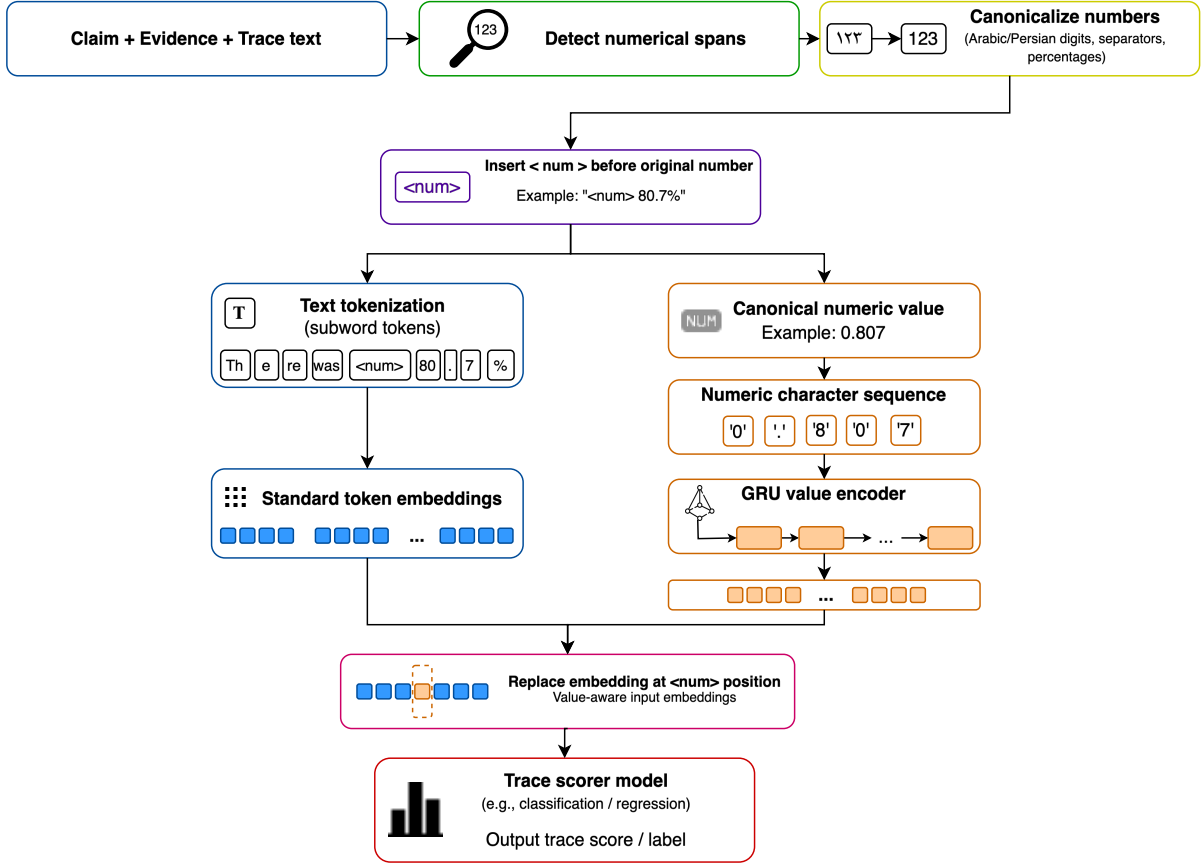


Fig. 7. Numeric embedding pipeline. Numerical spans are canonicalized, marked with `<num>`, encoded with a GRU value encoder, and inserted as value-aware embeddings before trace scoring.

preference pairs: the chosen output ranks gold-matching traces first and predicts the gold verdict, while rejected outputs are hard negatives (a non-matching trace placed first, swapped top traces, or an incorrect verdict). Because DPO optimizes preferences between complete generated rankings rather than per-trace scores, it inherits the order/format sensitivity of generative ranking and does not directly optimize Recall@5, which is precisely the limitation the trace scorer was designed to avoid.

3.8 Evaluation Protocol

In keeping with the foundational definition of consistency, we report task accuracy and language invariance *separately*, so that the relationship between them is an empirical finding rather than a definitional artifact [9, 28] (RQ2).

Task metrics. For each language we report Recall@5 and MRR@5 over gold-matching traces, macro-F1 of the predicted claim verdict, and the official average of macro-F1 and Recall@5; we also report the per-language

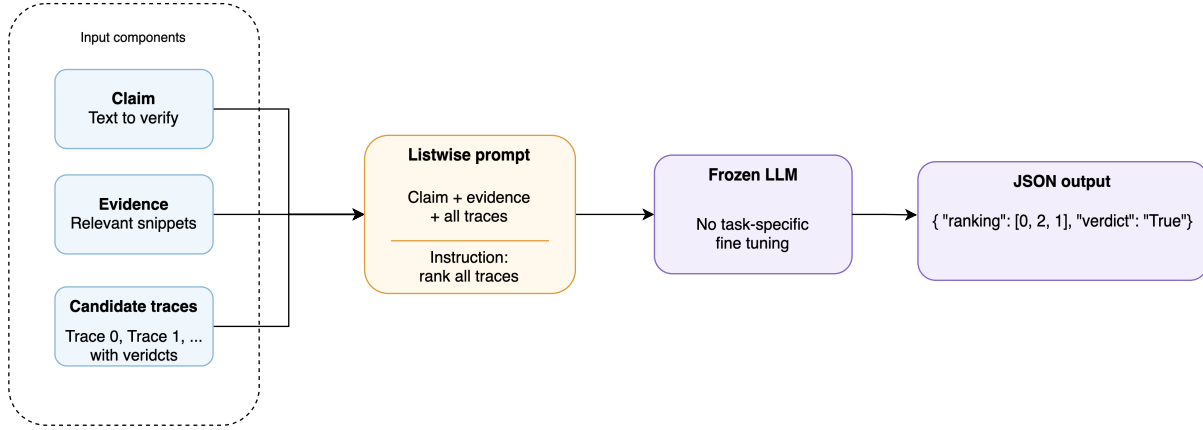


Fig. 8. Listwise prompted ranking setup. A frozen language model receives the claim, evidence, and all candidate traces in one prompt, then generates a complete ranking and final verdict.

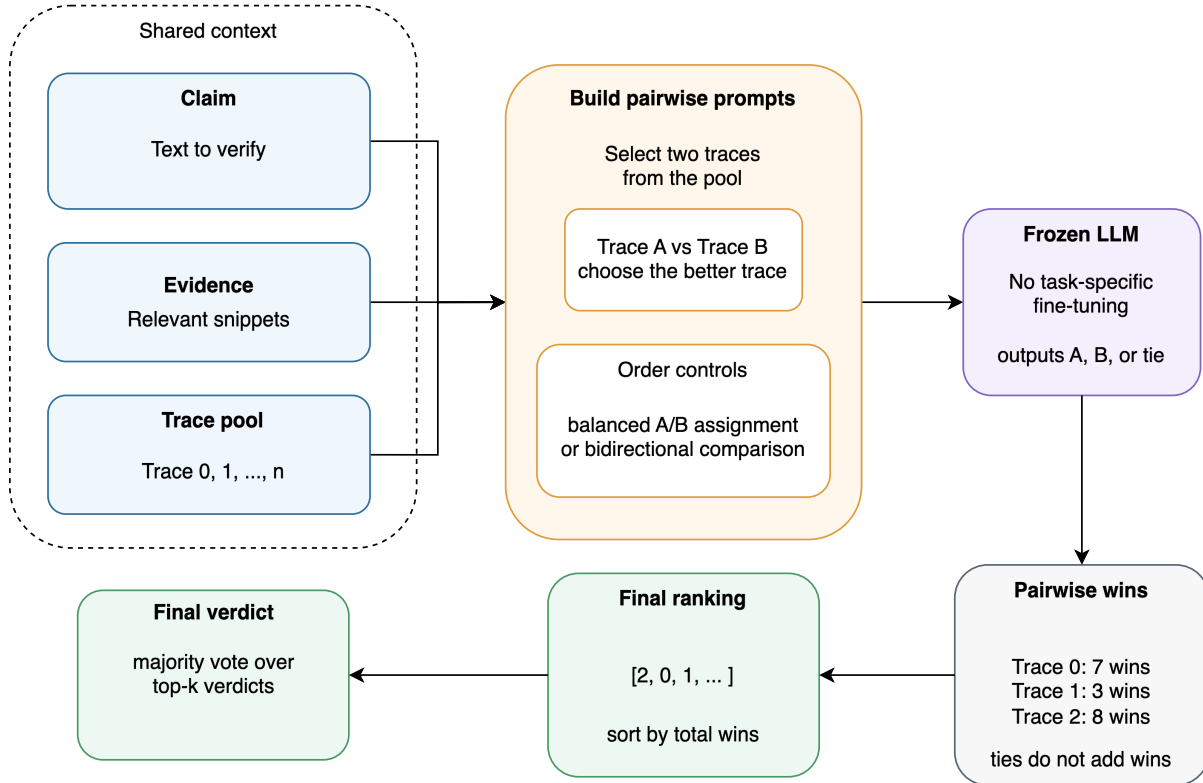


Fig. 9. Pairwise prompted ranking setup. A frozen language model compares two traces at a time, and accumulated pairwise wins are used to produce the final ranking.

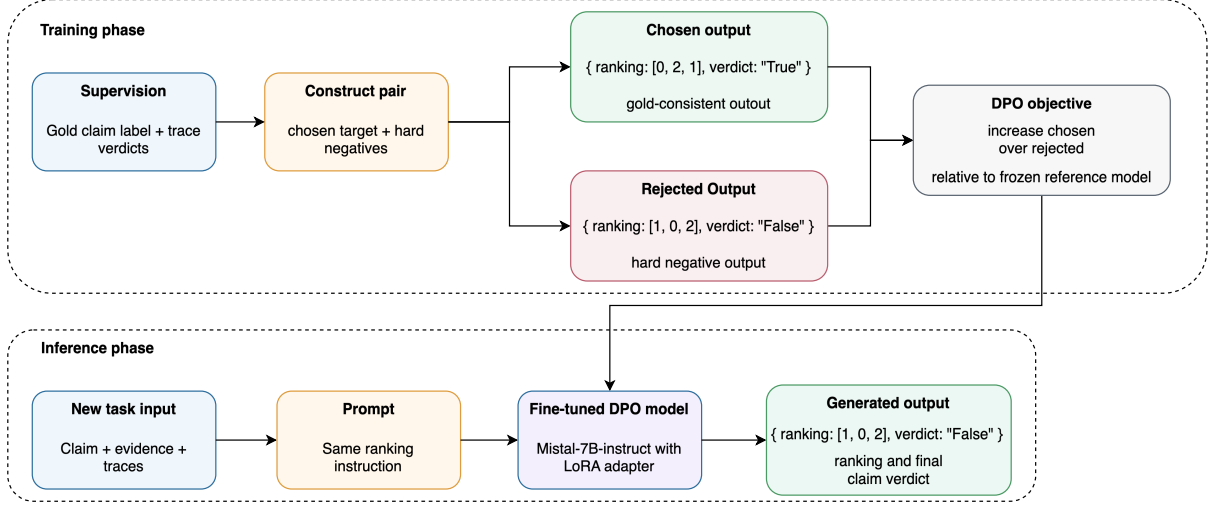


Fig. 10. DPO ranking setup. Chosen and rejected JSON outputs are automatically constructed from the gold label and trace verdicts, then used to train the model to prefer gold-consistent rankings.

Recall@5 upper bound so that absolute numbers can be read against attainable maxima. Multilingual runs additionally report each metric per language and as a macro-average across languages.

Verdict invariance. Treating the language-conditioned verdicts $\{\hat{y}_\ell(c)\}_{\ell \in \mathcal{L}}$ for a claim as the judgments of several raters, we measure their agreement with: the fraction of claims on which all languages agree (complete agreement); Fleiss’ κ across all languages [11]; pairwise Cohen’s κ [5]; and Krippendorff’s α , which tolerates missing variants [21]. These are computed over the aligned groups so that only genuinely parallel claims are compared.

Ranking invariance. As a finer diagnostic we quantify agreement among the language-conditioned trace orderings $\{r_\ell(c)\}$ using mean pairwise Kendall’s τ and Spearman’s ρ [19, 42], computed over the trace indices common to both orderings, together with the top- k Jaccard overlap of the ranked trace sets. The same machinery also supports cross-variant checks—e.g. original-language versus translated, or the claim/trace-language mismatched validation sets—which isolate whether an inconsistency is driven by the language of the claim, the language of the traces, or translation noise.

Training-time invariance. Finally, the multilingual evaluator reports $\mathcal{L}_{\text{inv}}$ itself—the mean standardized score MSE of Equation (5), overall and per language pair—on held-out data. This is the quantity the objective directly minimizes, and tracking it alongside the behavioral κ/τ statistics lets us check whether reducing score-level disagreement actually translates into the verdict- and ranking-level invariance we ultimately care about, and whether it does so without sacrificing accuracy [16].

3.9 Experimental Configuration

Training details. Unless otherwise noted, trace scorers are trained for five epochs with the AdamW optimizer, learning rate 1×10^{-4} , a warm-up ratio of 0.03 for the learning-rate schedule, per-device batch size up to 8 with gradient accumulation, and bf16 compute. Model selection uses early stopping with a small patience on a validation metric—positive-class F1 for single-language runs and the macro-F1/Recall@5 mean for multilingual

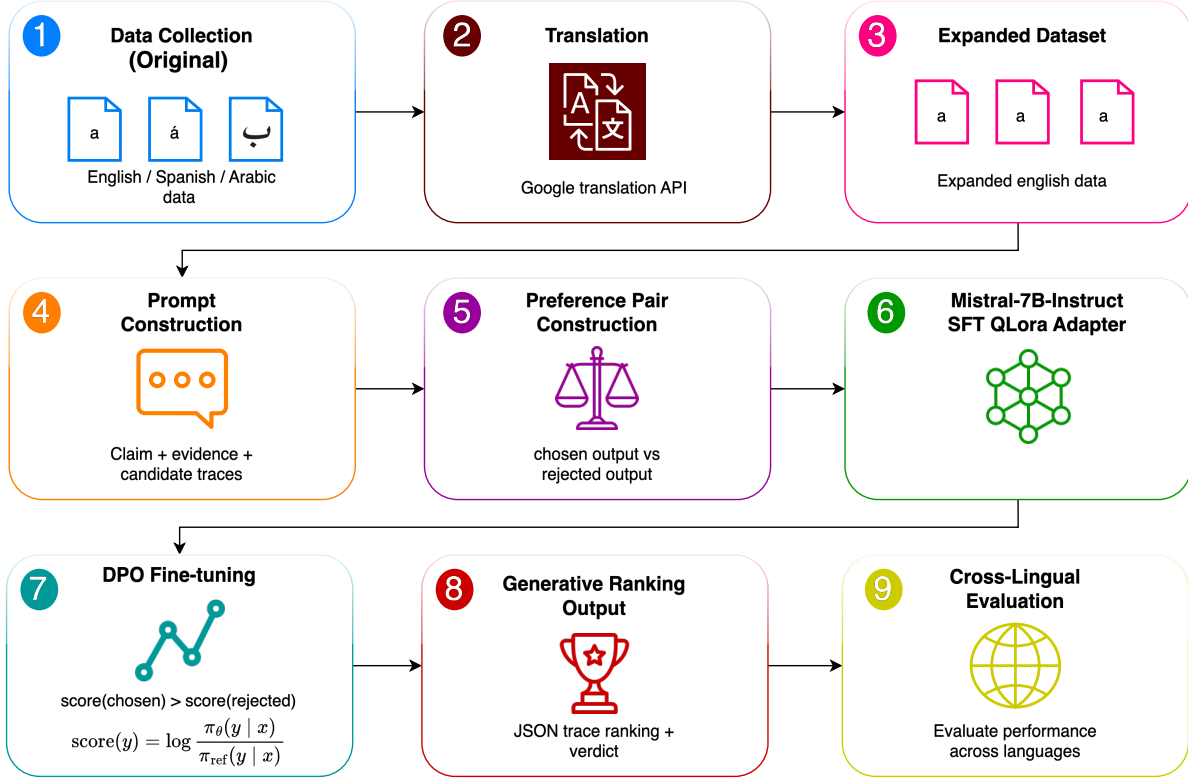


Fig. 11. DPO ranking pipeline. Translated multilingual data is converted into ranking prompts and chosen/rejected preference pairs for DPO fine-tuning, followed by cross-lingual evaluation.

runs—and the best adapter is saved at the evaluation step where that metric peaks rather than at the last step. For the language-invariance runs we set the invariance warm-up ratio $\rho = 0.2$ and sweep the target weight λ^* , always including $\lambda^* = 0$ as the no-invariance control. A *regularized* configuration increases adapter dropout to 0.10 and adds weight decay 0.01 to reduce overfitting on the long-context inputs. To account for run-to-run variance, every reported configuration is trained and evaluated with three random seeds, and we report the mean across the three runs. Experiment tracking and metric logging are handled with Weights and Biases.

Compute. Experiments run on NVIDIA A100 GPUs on the institutional cluster. The main trace-scorer runs use distributed data-parallel training across two GPUs via `torchrun`, while inference and smaller ablations use a single GPU. All LLM-based scorers use 4-bit base-model loading with QLoRA adapters and SDPA attention, with CUDA expandable memory segments enabled to accommodate the long claim–evidence–trace sequences.

Experiment matrix. The three backbones play different roles in the study. **Mistral-7B-Instruct-v0.3** was carried through the full set of ranking formulations during the shared task and is retained as a reference point rather than extended with the invariance objective. **Llama-3.1-8B-Instruct** is the primary backbone: it has completed both the ranking experiments (prompted baselines, DPO, and the multilingual trace scorers, including the long-context and regularized variants) and the language-invariance experiments, and serves as the reference against which the invariance objective is evaluated. We additionally treat plain-text numeric normalization as a

first-class setting on this backbone rather than a peripheral ablation. **Qwen3-8B** is evaluated under the matching set of conditions—the prompted baseline, the best ranking configuration identified for Llama (the multilingual, long-context, regularized trace scorer), the numeric-normalization setting, and the language-invariance objective. Evaluating all three backbones under the same conditions is what allows the thesis to separate effects of the invariance objective from idiosyncrasies of any single model, and to study how those effects vary across English, Spanish, and the lower-resource, different-script Arabic setting (RQ2, RQ3).

4 Results and Discussion

This chapter reports and interprets the experimental results. We first describe the evaluation setup and the data-hygiene decisions that determine which runs enter the analysis (Section 4.1). We then present task performance (Section 4.2) and cross-lingual consistency (Section 4.3) separately, as the methodology requires, before bringing them together to answer whether optimizing for language invariance improves ranking (Section 4.4, RQ1). Section 4.5 examines where the cross-lingual differences actually live, Section 4.6 covers the invariance weight, numeric normalization, and backbone families, and Section 4.7 analyses the relationship between invariance and reasoning quality (RQ2). Section 4.8 states the threats to validity, including a context-budget confound that qualifies the accuracy comparison, and Section 4.9 summarizes the findings against the three research questions (RQ3).

4.1 Experimental Scope and Data Hygiene

All results are computed on the aligned trilingual validation set described in Section 3.2: 4,233 claims present in English, Spanish, and Arabic with parallel trace sets, evaluated per language and, for consistency measures, across the three language-conditioned outputs of the same claim. Every reported configuration was trained and evaluated with three random seeds and we report the mean. In total we evaluate 32 configurations spanning prompted baselines, the earlier trace scorers, standard multilingual fine-tuning, and the invariance-trained models introduced in this thesis.

Excluded runs. Four of the 32 configurations are excluded from every statistic reported below. The rows *Mistral*, *English DPO*, *Llama*, *inv. $\lambda^*=0.05$* , and *Qwen, prompted* carry scores that are identical to one another to four decimal places and identical across all three languages, together with cross-lingual agreement of exactly 1.000 on every measure. Three distinct systems cannot plausibly produce byte-identical scores; these rows are duplicated or failed runs rather than results, and reporting them would contaminate every group mean. We exclude them and report the analysis over the remaining 28 configurations.

The fourth exclusion is different in kind and worth stating explicitly, because it illustrates the central measurement principle of this thesis. The *random ranking* baseline also achieves perfect cross-lingual agreement—Fleiss $\kappa = 1.000$ and Kendall $\tau = 1.000$ on every language pair. This is not an artefact: the baseline ranks traces in a fixed seeded permutation of trace indices that does not depend on the input at all, so it produces literally the same ranking in every language and is *invariant by construction*. It is also useless, sitting at the bottom of every accuracy measure. A system can therefore be perfectly language-invariant and still carry no information, which is precisely why invariance must be reported alongside, and never instead of, task accuracy [9, 28]. We exclude the random baseline from correlation statistics for the same reason—it would dominate them while being uninformative—but retain it in Table 1 as the accuracy floor.

Achievable ceiling. As established in Section 3.1, Recall@5 is structurally capped. On this validation set the upper bound is 0.3005, and 578 of the 4,233 claims (13.7%) have no candidate trace whose verdict matches the gold label, contributing zero Recall@5 regardless of the ranking. All Recall@5 figures below should be read against this ceiling rather than against 1.0.

4.2 Task Performance

Table 1 condenses the results: for each training group it reports the group mean together with that group’s most accurate and most cross-lingually stable member, placing accuracy and consistency side by side. Per-model results for all 28 configurations are given in Appendix A (Tables 2 and 3). Several observations follow.

Table 1. Condensed results. For each training group we report the group mean over its n configurations, followed by its most accurate (acc.) and, where different, its most cross-lingually stable (stab.) member. Accuracy and consistency are shown together so the trade-off is visible in one place: the right-hand block gives Fleiss κ over the three language-conditioned verdicts and mean pairwise Kendall τ between the English and Arabic rankings—what the invariance objective targets. Bold marks the best value in each column over all 28 configurations. Full per-model results are given in Tables 2 and 3.

Configuration	n	Accuracy (macro-F1)				Consistency	
		EN	ES	AR	Mean	κ	τ
<i>Invariance-trained (this thesis)</i>	13	0.6183	0.6176	0.6085	0.6148	0.9443	0.5667
Qwen, inv. $\lambda^*=0.1$ (acc.)		0.6192	0.6197	0.6157	0.6182	0.9479	0.5820
Llama, inv. $\lambda^*=0.3$ (stab.)		0.6165	0.6150	0.6034	0.6116	0.9428	0.6440
<i>Standard multilingual fine-tuning</i>	9	0.6158	0.6224	0.5847	0.6076	0.9028	0.1308
Llama, standard (acc.)		0.6256	0.6320	0.6105	0.6227	0.8810	0.0863
Qwen, num. norm. (stab.)		0.6221	0.6258	0.5810	0.6096	0.9272	0.2496
<i>Earlier trace scorers</i>	4	0.5861	0.5878	0.5851	0.5863	0.8596	0.0578
Mistral, multiling. trace scorer (10 ep.) (acc.)		0.5967	0.5932	0.5828	0.5909	0.9040	0.0955
Mistral, numeric trace scorer (stab.)		0.5826	0.5693	0.5938	0.5819	0.8643	0.1075
<i>Prompted baselines</i>	2	0.5761	0.5756	0.5755	0.5757	0.9108	0.2185
Mistral, prompted (zero-shot) (acc.)		0.5771	0.5795	0.5718	0.5761	0.9168	0.3674
<i>By backbone family (all settings pooled)</i>							
Llama	13	0.6139	0.6164	0.5948	0.6083	0.9195	0.3161
Qwen	10	0.6175	0.6189	0.6020	0.6128	0.9353	0.4505
Mistral	5	0.5843	0.5861	0.5825	0.5843	0.8711	0.1197
<i>Random ranking (floor; invariant by constr.)</i>	1	0.5820	0.5820	0.5820	0.5820	1.0000	1.0000

Every trained system beats the baselines, but the margins are narrow. The random ranking baseline reaches 0.2181 Recall@5 and 0.5820 macro-F1 in English, and the zero-shot prompted baselines are barely distinguishable from it (0.2234 and 0.5751 for the Llama prompted baseline). This reproduces, on a larger and fully parallel evaluation set, the finding from our shared-task system that zero-shot generative ranking adds little over chance in this setting. The fine-tuned trace scorers improve substantially on macro-F1 (up to 0.6263 in English) and modestly on Recall@5 (up to 0.2710).

Ranking headroom is nearly exhausted; verdict quality is not. The best Recall@5 of 0.2710 corresponds to roughly 90% of the achievable ceiling of 0.3005. In other words, the ranking component of the task is close to saturated for this candidate-trace set: further gains cannot come from better ranking, only from candidate sets that contain more gold-matching traces. Macro-F1, by contrast, ranges from 0.5715 to 0.6320 across configurations and remains the component where modelling choices still matter. This asymmetry should be kept in mind when reading the official combined score, which averages the two: it is increasingly dominated by the verdict term.

Metric redundancy. Across the 28 configurations, Recall@5, Precision@5, MRR@5, macro-F1, and the per-class F1 scores for True and False are almost collinear, correlating at Pearson r between 0.92 and 0.99. The single exception is F1 on the minority Conflicting class, which correlates with the others at only $r = 0.61$ –0.85.

Most of the headline metrics therefore restate the same ordering of systems, and Conflicting is the class that actually discriminates between them. We return to this class in Section 4.5, where it turns out to carry the entire cross-lingual effect.

4.3 Cross-Lingual Consistency

The right-hand block of Table 1 reports the consistency measures, computed independently of accuracy (per-model values in Table 3). The central result of this section is that the two families of consistency measure behave completely differently, and that only one of them discriminates between systems.

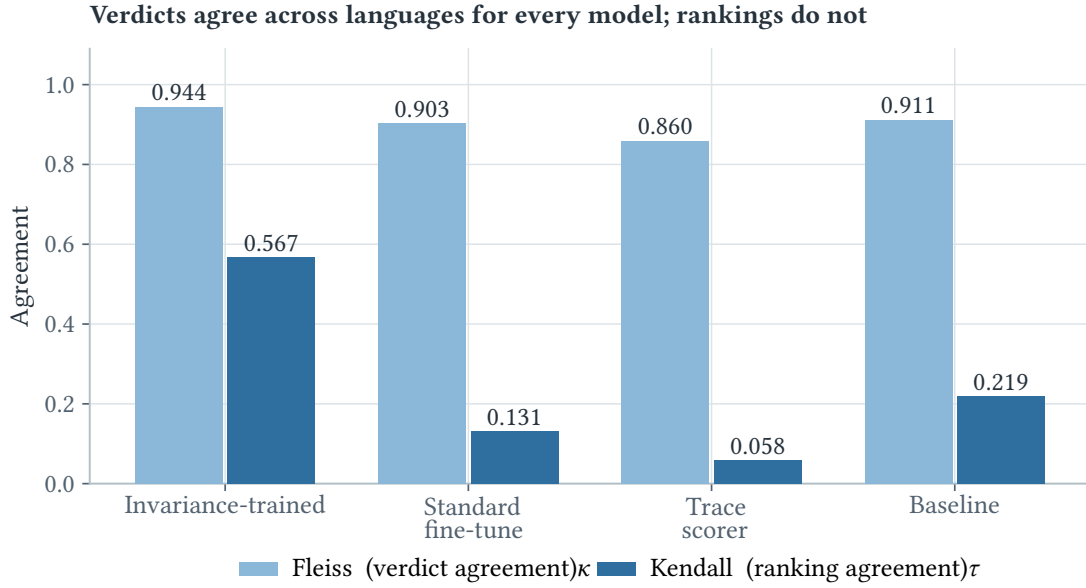


Fig. 12. Verdict agreement versus ranking agreement, averaged by training type. Fleiss κ over the three language-conditioned verdicts is high (0.86–0.94) for every group and separates them weakly. Mean pairwise Kendall τ between the language-conditioned trace rankings separates the same groups by almost an order of magnitude.

Verdicts agree across languages; rankings do not. As Figure 12 shows, Fleiss κ over the language-conditioned verdicts is uniformly high and occupies a narrow band: 0.8596 for the earlier trace scorers, 0.9028 for standard fine-tuning, and 0.9443 for invariance-trained models. Judged on verdict agreement alone, every system looks strongly language-invariant. Kendall τ between the language-conditioned trace *rankings* tells a very different story: on the English–Arabic pair it spans 0.012 to 0.644 across individual configurations and 0.0578 to 0.5667 across groups, and on the English–Spanish pair the weakest configuration is very slightly negative (−0.008), i.e. no better than an unrelated ordering.

The interpretation is that standard multilingual fine-tuning produces models that reach the same verdict in all three languages while justifying it with almost entirely different evidence: the top-ranked traces in English and Arabic are close to uncorrelated ($\tau \approx 0.13$). Because the final verdict is a majority vote over the top five traces, a model can arrive at a stable label through unstable and largely disjoint evidence. Verdict-level agreement statistics therefore substantially overstate cross-lingual robustness, and ranking-level agreement is the measure

that exposes the difference. This is a methodological finding in its own right: had we followed the common practice of reporting only κ over final answers, we would have concluded that all our systems were already language-invariant.

4.4 Does Invariance Training Improve Ranking? (RQ1)

Figure 13 places every configuration in the plane defined by mean macro-F1 (accuracy) and Kendall τ between English and Arabic rankings (stability). The picture is unusually clean.

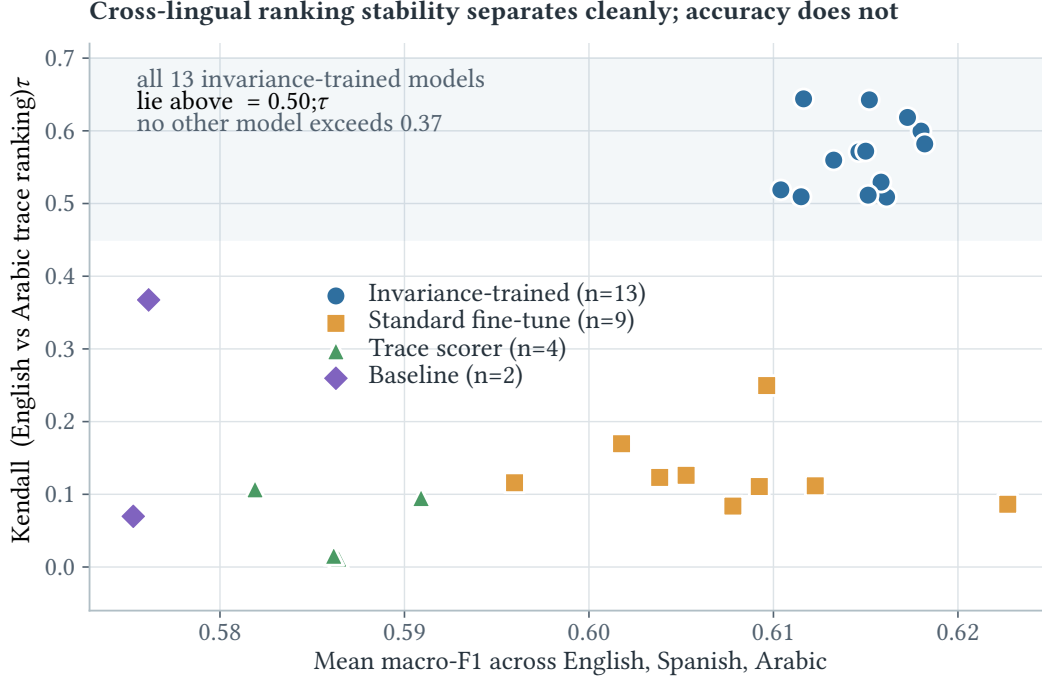


Fig. 13. Each point is one configuration ($n = 28$). The vertical axis is the mean pairwise Kendall τ between the English and Arabic trace rankings; the horizontal axis is macro-F1 averaged over the three languages. The invariance-trained population is fully separated from every other configuration on stability, while the accuracy ranges overlap.

The stability effect is large and fully separated. Table 1 summarizes the group-level picture. The 13 invariance-trained configurations reach a mean Kendall τ of 0.5667 between English and Arabic, against 0.1308 for standard fine-tuning, 0.0578 for the earlier trace scorers, and 0.2185 for the baselines. The two populations do not overlap at all: the *worst* invariance-trained configuration ($\tau = 0.5089$) exceeds the *best* non-invariance configuration ($\tau = 0.3674$) by a wide margin. The effect size is Cohen’s $d = 6.09$ (Welch $t = 16.71$, $p < 10^{-13}$; Mann–Whitney $p < 10^{-5}$). Verdict agreement moves in the same direction, from Fleiss $\kappa = 0.8924$ to 0.9443 ($d = 2.20$, $p < 10^{-4}$). Complete separation of two populations is a stronger statement than any p -value computed from 28 points, and it is the claim we rest on.

The accuracy effect is positive but not the argument. Mean macro-F1 is also higher for invariance-trained models, 0.6148 against 0.5977 ($d = 1.61$, $p < 0.001$). However, unlike stability, the accuracy *ranges overlap*: the best

non-invariance configuration (*Llama, standard*, 0.6227) outperforms the best invariance-trained one (*Qwen, inv.* $\lambda^*=0.1$, 0.6182). If invariance training were judged on English macro-F1 alone it would read as a null result, with the best English score belonging to *Llama, reg.* (0.6263).

The honest answer to RQ1 is therefore conditional, and it is worth stating precisely. Optimizing for language invariance does *not* produce a better ranker in the sense of a higher peak score on any individual language. What it produces is a ranker whose ordering of reasoning traces is stable under translation—a roughly fourfold improvement in cross-lingual ranking agreement—at no cost to accuracy, and with a small positive accuracy effect on average. Since the property being optimized is cross-lingual consistency rather than per-language quality, this is the outcome the objective was designed to produce; but it means the contribution should be claimed as robustness, not as state-of-the-art ranking performance.

4.5 Where the Cross-Lingual Differences Live

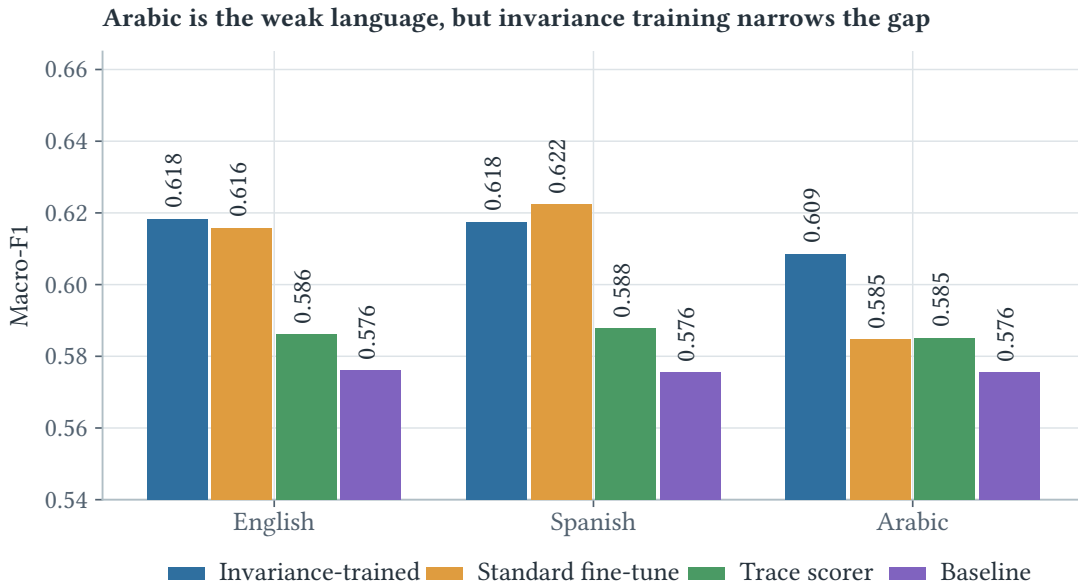


Fig. 14. Macro-F1 by language, averaged by training type. Spanish is on par with English for every group; Arabic is consistently the weakest language, and the gap is smallest for invariance-trained models.

Averaged over the 28 configurations, Spanish macro-F1 (0.6118) is marginally *above* English (0.6099), while Arabic (0.5951) sits 0.0147 below English, a relative drop of about 2.4%. Recall@5 follows the same pattern (0.2579 / 0.2620 / 0.2548). The multilingual difficulty in this task is thus almost entirely an Arabic effect, consistent with the expectation from Section 2.7 that a different numeral script and lower pre-training representation would make Arabic the hard case [28, 34].

Figure 15 localizes the effect. The aggregate Arabic drop is not spread evenly over the label space. F1 on False and True is almost flat across languages, losing only 0.0026 and 0.0027 respectively between English and Arabic, whereas F1 on the minority Conflicting class falls from 0.2625 (English) and 0.2671 (Spanish) to 0.2236 (Arabic)—a drop roughly fourteen times larger than for either majority class. Broken down by training type the contrast is sharper still. Standard fine-tunes drop from 0.270 in English to 0.178 in Arabic—a collapse of roughly a

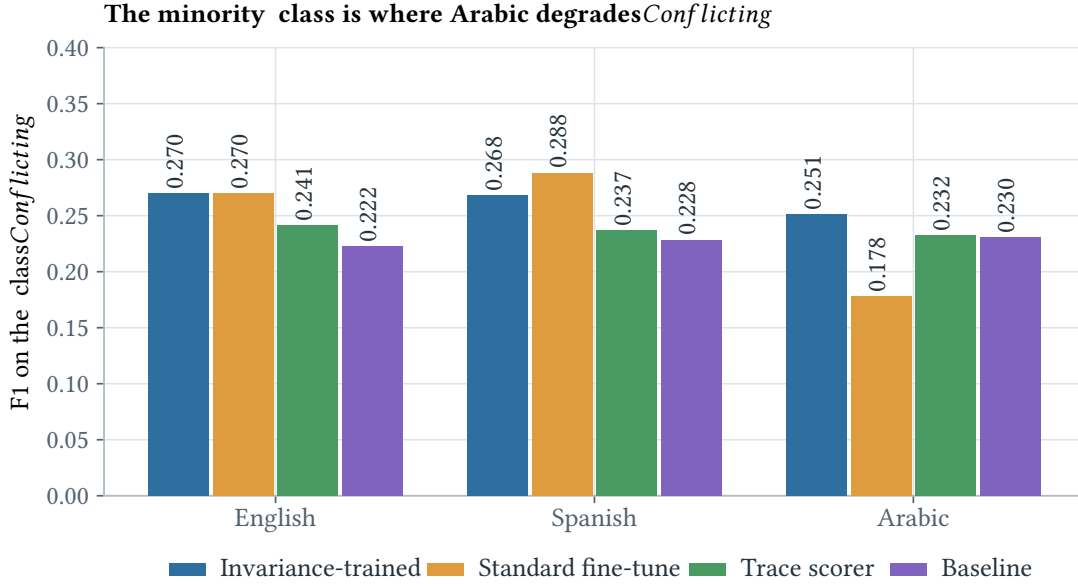


Fig. 15. F1 on the minority Conflicting class by language. The Arabic degradation seen in Figure 14 is concentrated almost entirely in this class: standard fine-tunes fall from 0.270 in English to 0.178 in Arabic, while invariance-trained models hold at 0.251.

third—whereas invariance-trained models fall only from 0.270 to 0.251. Conflicting is both the rarest class (427 of 4,233 claims, 10.1%) and the one that requires genuinely weighing contradictory quantitative evidence rather than matching a single number, so it is the class where unstable evidence selection is most damaging. The class-level view thus explains the aggregate: invariance training helps Arabic specifically by protecting the hardest and rarest decision.

A caution on the aggregate Arabic gap. The mean English-to-Arabic macro-F1 drop is -0.0098 for invariance-trained models against -0.0311 for standard fine-tuning, a difference of roughly threefold in the expected direction. We note, however, that this particular comparison does *not* reach statistical significance (Welch $p = 0.089$; Mann-Whitney $p = 0.231$), because the per-configuration variance in the gap is large relative to n . The direction is consistent across families and settings and is corroborated by the class-level analysis above, but it should be reported as a consistent trend rather than an established effect. This contrasts with the ranking-stability result, which is unambiguous.

4.6 Ablations

The invariance weight. Figure 16 sweeps the target invariance weight λ^* over $\{0.1, 0.2, 0.3\}$ for both backbones. Cross-lingual stability increases monotonically with λ^* for both families—Llama from 0.5711 to 0.6440 and Qwen from 0.5820 to 0.6426—which is the expected behaviour of a regularizer whose strength is being increased. Mean macro-F1 does not follow: it peaks at $\lambda^* = 0.2$ for Llama (0.6180, against 0.6146 and 0.6116 at 0.1 and 0.3) and declines slightly but monotonically for Qwen (0.6182 $\rightarrow$ 0.6173 $\rightarrow$ 0.6152). The sweep therefore exposes a mild trade-off rather than a free lunch, and $\lambda^* = 0.2$ is the setting that balances the two. It should be stressed that the

entire swept range remains far above the non-invariance population on stability, so the choice of λ^* within this range is much less consequential than the decision to use the objective at all.

Stability rises monotonically with ; accuracy peaks at $\lambda^* = 0.2$

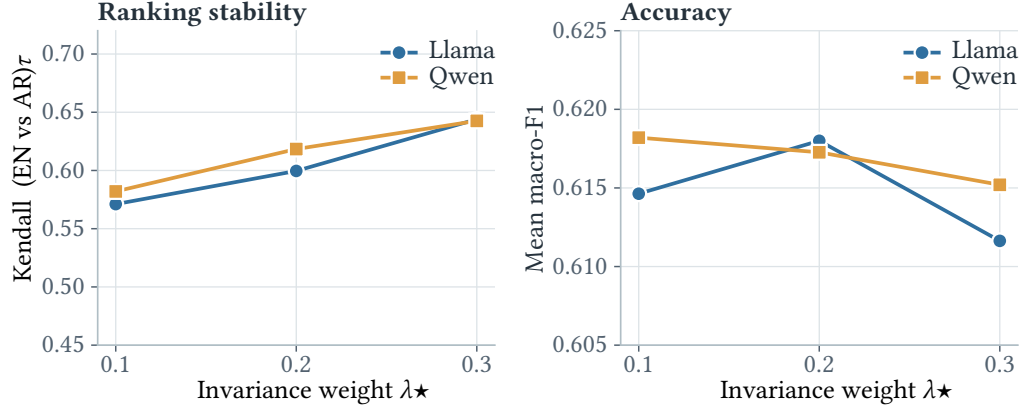


Fig. 16. Effect of the target invariance weight λ^* . Ranking stability rises monotonically with λ^* for both backbones, while mean macro-F1 peaks at $\lambda^* = 0.2$ and then declines for Llama.

Numeric normalization does not combine with invariance training. Plain-text numeric normalization was carried over from the shared-task system as a complementary lever for the numerical dimension of the task (Section 3.6). Within the invariance-trained group it does not help. Configurations combining the two reach $\tau = 0.5121$ and mean macro-F1 0.6133, against $\tau = 0.5910$ and 0.6154 for invariance training alone: adding numeric normalization costs 0.079 Kendall τ and buys no accuracy, and the same pattern holds in Arabic (0.6028 against 0.6111). Outside the invariance group the picture is different—among standard fine-tunes, numeric normalization is mildly positive in English and Spanish—so the two interventions appear to be partially redundant rather than individually useless. A plausible reading is that both mechanisms address the same underlying problem: numeric normalization canonicalizes numerals so that they are represented identically across scripts, which is one of the specific ways the invariance objective already forces scores to agree. Once the score-level constraint is imposed, restating canonical values in the input adds input length without adding signal. On this evidence, the two should not be combined.

Backbone families. Pooling all settings, Qwen3-8B and Llama-3.1-8B are level on accuracy (mean macro-F1 0.6128 against 0.6083) but Qwen is noticeably more stable cross-lingually ($\tau = 0.4505$ against 0.3161; Fleiss $\kappa = 0.9353$ against 0.9195). Mistral-7B trails on both dimensions (0.5843 and 0.1197), confirming the earlier decision to retain it as a reference point rather than extend it with the invariance objective. The strongest individual results are split: *Llama, reg.* attains the best English macro-F1 (0.6263) and Recall@5 (0.2710), *Llama, inv. $\lambda^* = 0.2$* the best Arabic macro-F1 (0.6161), and *Llama, inv. $\lambda^* = 0.3$* the best ranking stability ($\tau = 0.6440$). That no single configuration wins on every axis is itself consistent with the accuracy/stability distinction drawn in Section 4.4.

4.7 The Relationship Between Invariance and Reasoning Quality (RQ2)

RQ2 asks how language invariance relates to reasoning quality. Because the two were measured independently, the relationship can be examined empirically. Over the 28 configurations, mean macro-F1 correlates with Kendall τ at $r = 0.60$ and with Fleiss κ at $r = 0.57$; English Recall@5 correlates with τ at $r = 0.49$. The association is therefore positive and moderate: better models do tend to be more cross-lingually stable, as the thesis premise anticipated.

The association is, however, far from deterministic, and Figure 13 makes the residual visible. At essentially identical mean macro-F1 (around 0.61–0.62) the configurations span τ from 0.08 to 0.64—a range of more than 0.5 at fixed accuracy. Accuracy consequently predicts stability only weakly, and the two properties are separable in practice as well as in principle. This is the empirical confirmation, in a ranking setting, of the decoupling that the consistency literature established for factual prediction [9, 28]: one cannot infer that a more accurate multilingual model is a more consistent one, and improving accuracy is not a route to consistency.

A second relationship is worth recording. Fleiss κ and Kendall τ correlate at $r = 0.83$ across configurations, so the two consistency families are related—but, as Section 4.3 showed, κ is compressed into a narrow high band while τ is not. A high correlation between two measures does not make them interchangeable when one has far less dynamic range than the other.

4.8 Threats to Validity

Context-budget confound. The most important qualification concerns training context. The invariance-trained models were trained with a reduced context budget, because the aligned trilingual batching required for the objective multiplies the sequence count per optimization step and the available compute did not permit the full long-context configuration. Several of the standard fine-tunes, by contrast, used the full budget. Training condition and context budget are therefore partially entangled, and the comparison is not a controlled ablation.

Two considerations limit the damage. First, the direction of the confound is conservative: the invariance arm is the handicapped one, so its wins are lower bounds. Second, the size of the context effect can be estimated from configurations that were run at both budgets. Moving the same architectures from the reduced to the full budget changes English macro-F1 by +0.0052, Spanish by +0.0043, and Arabic by −0.0053, for a net change of +0.0014 on the three-language mean—smaller than the invariance effect and not even consistent in sign across languages. Restricting the comparison to the matched reduced-budget condition, the invariance-trained models still exceed the short-context fine-tunes by +0.0083 mean macro-F1, +0.0197 Arabic macro-F1, and a factor of 4.85 on Kendall τ . We therefore state the accuracy claim in the guarded form that the evidence supports: *invariance training at a reduced context budget matches or exceeds full-context fine-tuning on accuracy, and dominates it on cross-lingual stability*. The stability result is unaffected by the confound, since no plausible context effect of the magnitude estimated here could produce a fourfold change in τ .

Small comparison arms. The matched-budget comparison rests on only two short-context fine-tunes (one Llama, one Qwen), and the full-budget arm has seven configurations. The group means are correspondingly imprecise, and the Spanish result is an explicit exception to the general pattern: *Llama, short context* reaches 0.6220 Spanish macro-F1, above all 13 invariance-trained configurations. We do not claim a Spanish improvement. The defensible summary is parity on English and Spanish, a gain on Arabic, and a large gain on cross-lingual stability.

Single evaluation set and translated data. All results come from one validation set, and the Spanish and Arabic sides of the aligned data are machine-translated (Section 3.2). Translation artefacts are therefore shared between training and evaluation, and the measured cross-lingual consistency is consistency with respect to *translated*

inputs rather than natively authored ones. Consistency on native text may be lower. Additionally, no invariance-trained configuration was run at the full context budget; that single run would resolve the confound described above and is the most valuable next experiment.

Behavioural, not representational, evidence. Finally, all invariance measures reported here are behavioural: they compare outputs across languages. As noted in Section 2.6, agreement of outputs does not establish that the model represents the underlying knowledge in a shared way [16]. Our results show that the objective changes what the model *does* across languages; they do not establish what it has changed internally.

4.9 Summary of Findings

RQ1 – Does optimizing for language-invariant reasoning improve ranking performance? Partly, and in a specific sense. The score-level invariance objective does not raise peak per-language ranking scores; the best English macro-F1 and Recall@5 both belong to a standard fine-tune. It does produce a fourfold, fully separated improvement in the cross-lingual stability of the ranking ($\tau = 0.567$ against 0.131, $d = 6.09$) with a small positive effect on mean accuracy (0.6148 against 0.5977). Ranking quality measured per language is essentially unchanged; ranking *consistency* across languages is transformed.

RQ2 – What is the relationship between language invariance and reasoning quality? Positive but moderate and clearly non-deterministic. Accuracy correlates with ranking stability at $r = 0.60$ and with verdict agreement at $r = 0.57$, yet at fixed accuracy the observed stability spans a range of more than 0.5 in Kendall τ . Consistency is not a by-product of accuracy, and the two must be measured separately—a conclusion reinforced by the random baseline, which is perfectly invariant and entirely uninformative.

RQ3 – Can language-invariant training objectives lead to more robust multilingual reasoning models?

Yes, with the qualifications above. The gains are concentrated where robustness is most needed: in Arabic, the lowest-resource and different-script language, and specifically in the minority Conflicting class, where standard fine-tunes lose roughly a third of their F1 and invariance-trained models lose almost none. The effect holds across both Llama and Qwen backbones, indicating a property of the method rather than of a single model family, and it is obtained under a reduced training-context budget.

A final observation cuts across all three questions. Standard multilingual fine-tuning already produces models that agree on the verdict in every language; what it does not produce is models that agree on *why*. The contribution of the invariance objective is to align the evidence, not the label—and because that difference is invisible to verdict-level agreement statistics, it would have gone unmeasured under the evaluation protocol most commonly used in this literature.

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A Full Per-Model Results

Tables 2 and 3 give the complete per-model figures condensed in Table 1. All values are means over three random seeds on the aligned trilingual validation set of 4,233 parallel claims. The four degenerate configurations described in Section 4.1 are omitted.

Table 2. Per-model results on the aligned trilingual validation set (4,233 parallel claims per language). *Mean* is the average macro-F1 over the three languages; best value per column in bold. All values are means over three random seeds.

Model	Recall@5			Macro-F1			Mean
	EN	ES	AR	EN	ES	AR	
<i>Invariance-trained (this thesis)</i>							
Qwen, inv. $\lambda^*=0.1$	0.2663	0.2667	0.2622	0.6192	0.6197	0.6157	0.6182
Llama, inv. $\lambda^*=0.2$	0.2699	0.2694	0.2667	0.6211	0.6168	0.6161	0.6180
Qwen, inv. $\lambda^*=0.2$	0.2661	0.2670	0.2630	0.6176	0.6185	0.6157	0.6173
Llama, inv. + num. norm.	0.2697	0.2698	0.2618	0.6227	0.6205	0.6052	0.6161
Qwen, inv. $\lambda^*=0.05$	0.2659	0.2663	0.2627	0.6174	0.6186	0.6115	0.6158
Qwen, inv. $\lambda^*=0.3$	0.2646	0.2648	0.2606	0.6173	0.6185	0.6098	0.6152
Qwen, inv. + reg. + num. norm.	0.2648	0.2671	0.2552	0.6211	0.6206	0.6037	0.6151
Qwen, inv. + reg.	0.2677	0.2657	0.2627	0.6202	0.6158	0.6090	0.6150
Llama, inv. $\lambda^*=0.1$	0.2671	0.2681	0.2618	0.6170	0.6175	0.6094	0.6146
Llama, inv. + reg.	0.2692	0.2693	0.2627	0.6177	0.6132	0.6089	0.6133
Llama, inv. $\lambda^*=0.3$	0.2683	0.2684	0.2638	0.6165	0.6150	0.6034	0.6116
Llama, inv. + reg. + num. norm.	0.2689	0.2706	0.2620	0.6137	0.6190	0.6018	0.6115
Qwen, inv. + num. norm.	0.2663	0.2662	0.2574	0.6163	0.6146	0.6003	0.6104
<i>Standard multilingual fine-tuning</i>							
Llama, standard	0.2649	0.2704	0.2692	0.6256	0.6320	0.6105	0.6227
Llama, reg.	0.2710	0.2783	0.2673	0.6263	0.6297	0.5808	0.6123
Qwen, num. norm.	0.2702	0.2762	0.2654	0.6221	0.6258	0.5810	0.6096
Qwen, short context	0.2642	0.2672	0.2649	0.6101	0.6161	0.6015	0.6092
Llama, num. norm.	0.2640	0.2732	0.2664	0.6154	0.6226	0.5854	0.6078
Llama, reg. + num. norm.	0.2583	0.2723	0.2649	0.6072	0.6261	0.5825	0.6053
Llama, short context	0.2632	0.2711	0.2583	0.6134	0.6220	0.5761	0.6038
Qwen, standard	0.2671	0.2737	0.2656	0.6135	0.6203	0.5715	0.6018
Llama, BCE head	0.2684	0.2729	0.2654	0.6085	0.6066	0.5728	0.5960
<i>Earlier trace scorers</i>							
Mistral, multiling. trace scorer (10 ep.)	0.2566	0.2549	0.2520	0.5967	0.5932	0.5828	0.5909
Mistral, trace scorer	0.2196	0.2490	0.2171	0.5847	0.5971	0.5775	0.5864
Mistral, multiling. trace scorer	0.2140	0.2430	0.2185	0.5806	0.5915	0.5864	0.5862
Mistral, numeric trace scorer	0.2184	0.2155	0.2228	0.5826	0.5693	0.5938	0.5819
<i>Baselines</i>							
Mistral, prompted (zero-shot)	0.2225	0.2173	0.2129	0.5771	0.5795	0.5718	0.5761
Llama, prompted (zero-shot)	0.2234	0.2222	0.2219	0.5751	0.5716	0.5792	0.5753
Recall@5 upper bound	0.3005			—			

Table 3. Cross-lingual consistency, measured separately from task accuracy. *Complete* is the fraction of claims on which all three languages produce the same verdict. Ranking agreement is the mean pairwise Kendall τ between the language-conditioned trace orderings. Best per column in bold.

Model	Verdict agreement		Ranking agreement (Kendall τ)		
	Complete	Fleiss κ	EN-ES	EN-AR	ES-AR
<i>Invariance-trained (this thesis)</i>					
Llama, inv. $\lambda^*=0.3$	0.9631	0.9428	0.8268	0.6440	0.6369
Qwen, inv. $\lambda^*=0.3$	0.9754	0.9623	0.8226	0.6426	0.6354
Qwen, inv. $\lambda^*=0.2$	0.9655	0.9470	0.7922	0.6184	0.6023
Llama, inv. $\lambda^*=0.2$	0.9657	0.9471	0.8316	0.5996	0.5941
Qwen, inv. $\lambda^*=0.1$	0.9648	0.9479	0.7680	0.5820	0.5671
Qwen, inv. + reg.	0.9627	0.9424	0.7396	0.5720	0.5411
Llama, inv. $\lambda^*=0.1$	0.9624	0.9425	0.7909	0.5711	0.5527
Llama, inv. + reg.	0.9634	0.9441	0.7830	0.5597	0.5472
Qwen, inv. $\lambda^*=0.05$	0.9646	0.9460	0.7274	0.5294	0.5111
Qwen, inv. + num. norm.	0.9622	0.9420	0.7479	0.5188	0.5138
Qwen, inv. + reg. + num. norm.	0.9587	0.9379	0.7473	0.5115	0.5060
Llama, inv. + reg. + num. norm.	0.9617	0.9394	0.7762	0.5093	0.4944
Llama, inv. + num. norm.	0.9579	0.9347	0.7666	0.5089	0.4998
<i>Standard multilingual fine-tuning</i>					
Qwen, num. norm.	0.9549	0.9272	0.3106	0.2496	0.2210
Qwen, standard	0.9407	0.9053	0.2683	0.1696	0.1496
Llama, reg. + num. norm.	0.9287	0.8872	0.1976	0.1261	0.1218
Llama, short context	0.9185	0.8738	0.2377	0.1232	0.1012
Llama, BCE head	0.9707	0.9519	0.1245	0.1159	0.1293
Llama, reg.	0.9431	0.9095	0.1912	0.1117	0.1146
Qwen, short context	0.9320	0.8948	0.1628	0.1107	0.0672
Llama, standard	0.9206	0.8810	0.2079	0.0863	0.1010
Llama, num. norm.	0.9341	0.8949	0.1732	0.0838	0.1456
<i>Earlier trace scorers</i>					
Mistral, numeric trace scorer	0.9022	0.8643	0.0435	0.1075	0.0933
Mistral, multiling. trace scorer (10 ep.)	0.9398	0.9040	0.0886	0.0955	0.0855
Mistral, multiling. trace scorer	0.8802	0.8298	0.0181	0.0161	0.0137
Mistral, trace scorer	0.8885	0.8404	-0.0084	0.0122	0.0179
<i>Baselines</i>					
Mistral, prompted (zero-shot)	0.9395	0.9168	0.6754	0.3674	0.4023
Llama, prompted (zero-shot)	0.9320	0.9049	0.1242	0.0697	0.3416